AoM -1

Chapter 2 Atomic Structure

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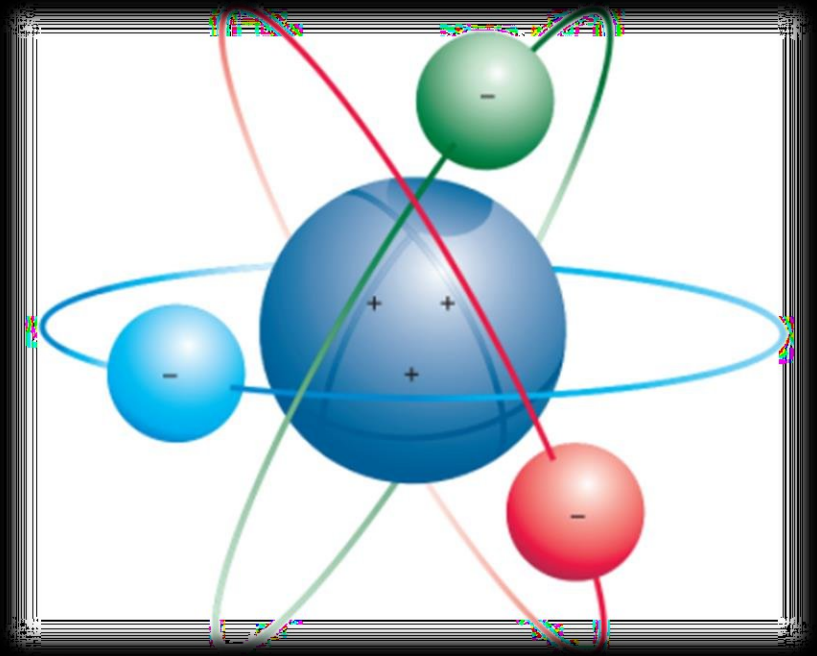
Based on : Patricia AOM-1 Lecture for UFAZ 2018/2019
And Manning, Phillip. Atoms, Molecules, and Compounds. Chelsea House Publishers, 2008.

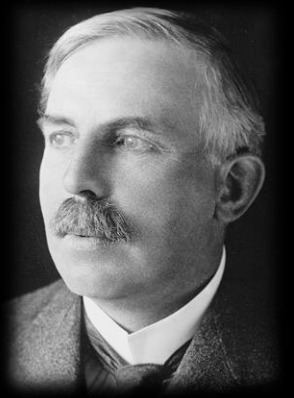
Summary

- Chapter 1 Basic Concepts and Definitions
- Chapter 2 Atomic Structure, Hydrogen Atom and Hydrogenoids
- Chapter 3 Polyelectronic Atoms
- Chapter 4 Periodic Table
- Chapter 5 Lewis Structure

Last Week

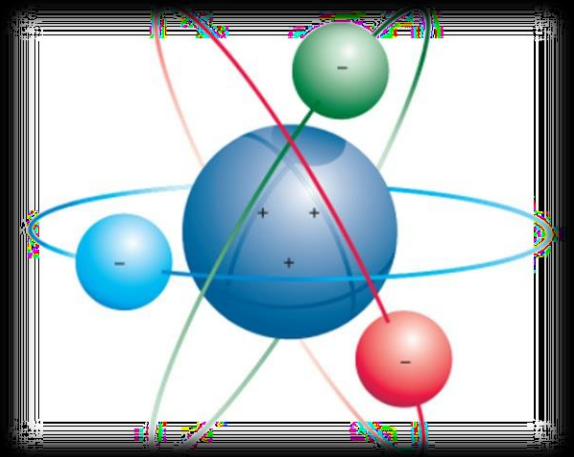
The orbital model of Rutherford





A positive charge concentrated in a small space called the nucleus, with even smaller negative charge particle circling around it

The Limitations of Rutherford's Atomic Model

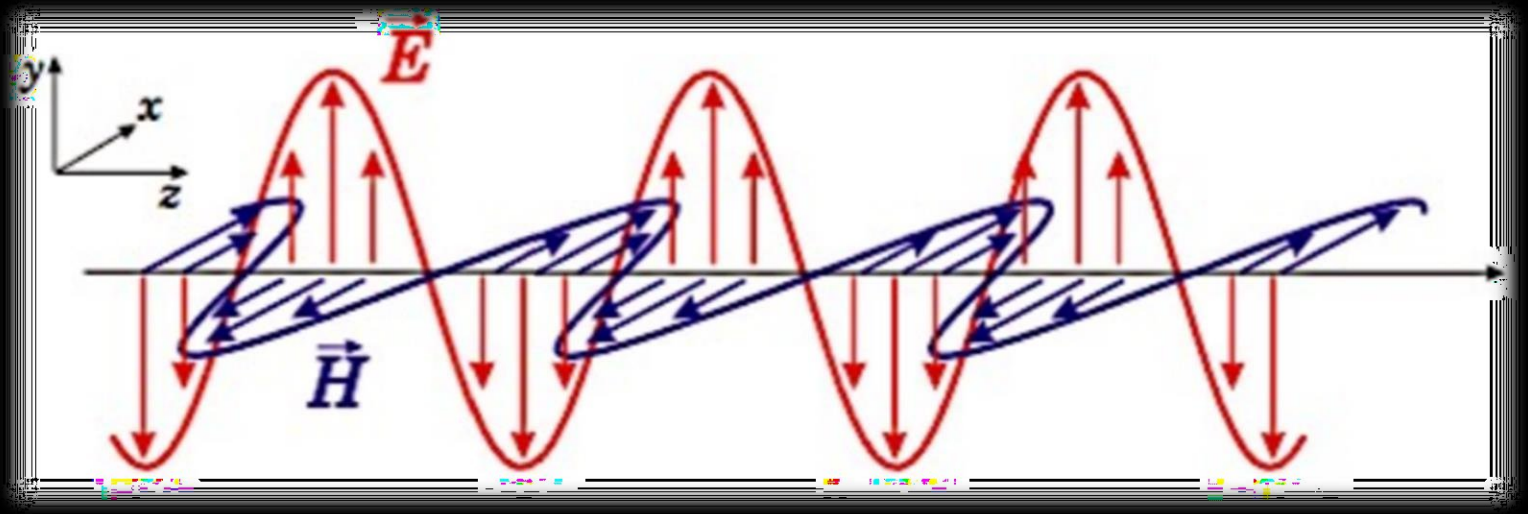


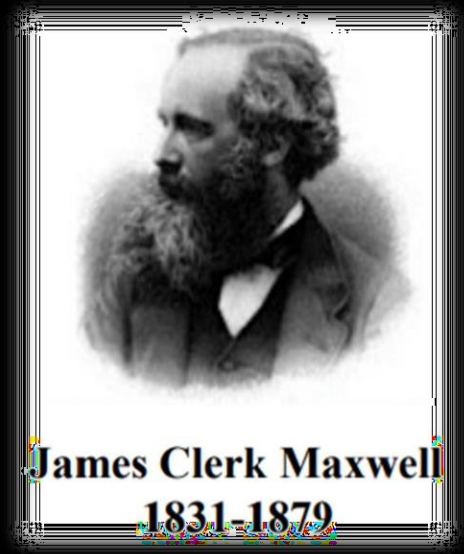
- It explained only a few properties of atoms.
- It could not explain the chemical properties of elements
- According to Maxwell's study, electron moving around the nucleus should emit an electromagnetic wave.
- Electron should then move with a helical movement and collide with the Nucleus. But the atom is stable.
- The atomic spectroscopy will determine more in detail the atomic structure

Nature of Light

By the year 1900, there was enough experimental evidence to convince scientists that light consisted of waves.

Light is a form of electromagnetic radiation, which consists of oscillating (time-varying) electric and magnetic fields that travel through empty space at about $c = 3.10^8\ m \cdot s^{-1}$



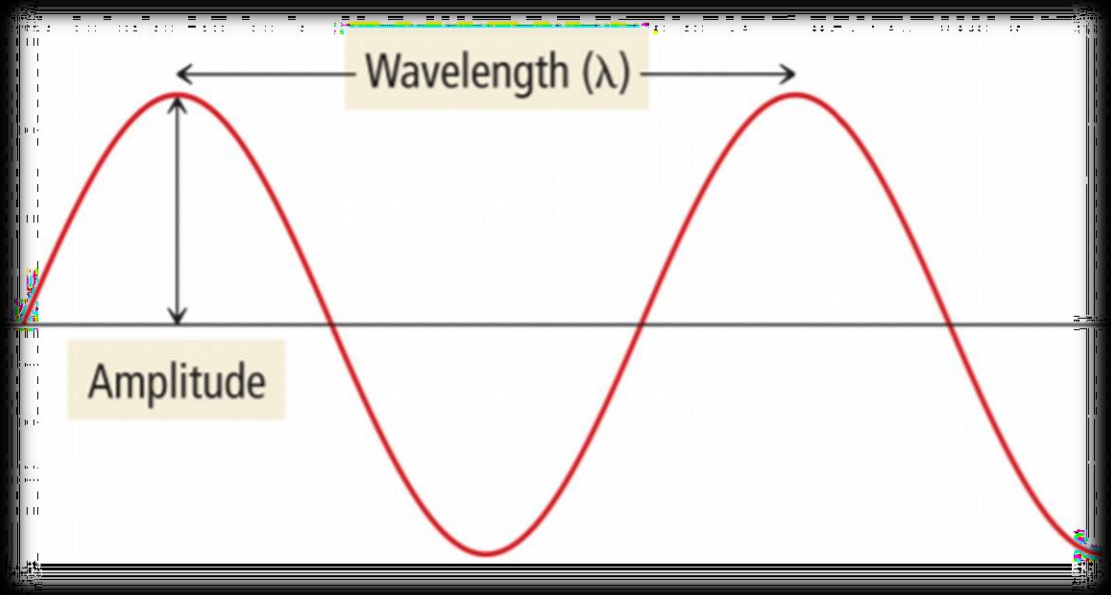


Nature of Light

The **amplitude** is the height of the wave.

The **wavelength (λ)** is a measure of the distance covered by the wave. **Its unit is the meter (m)**

The **frequency (ν)**, is the number of wave cycles to pass a given point per unit of time. **Its unit is the s^{-1} or Hertz (Hz)**

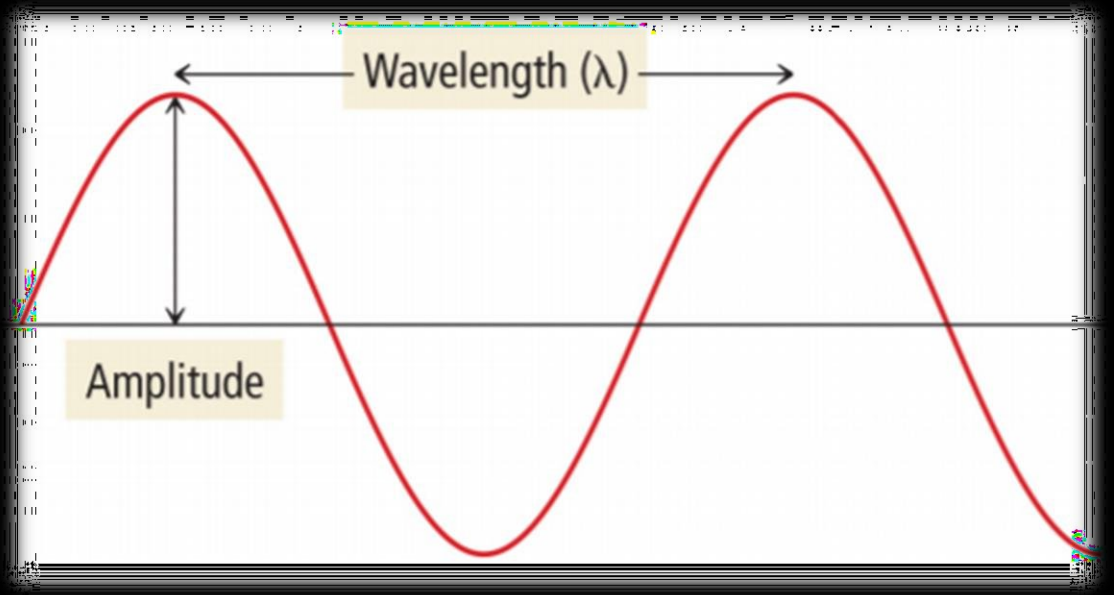


Nature of Light

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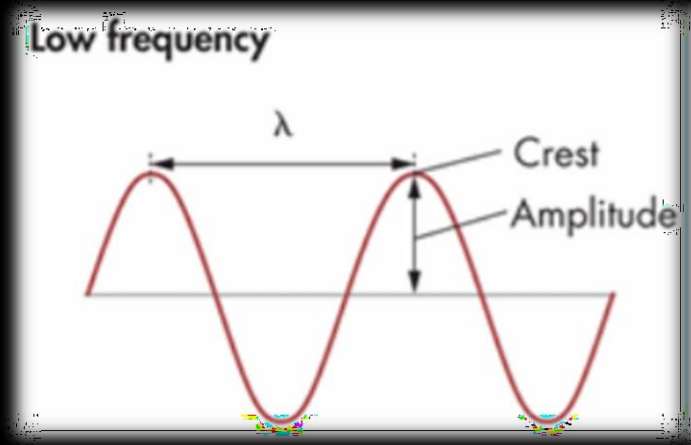
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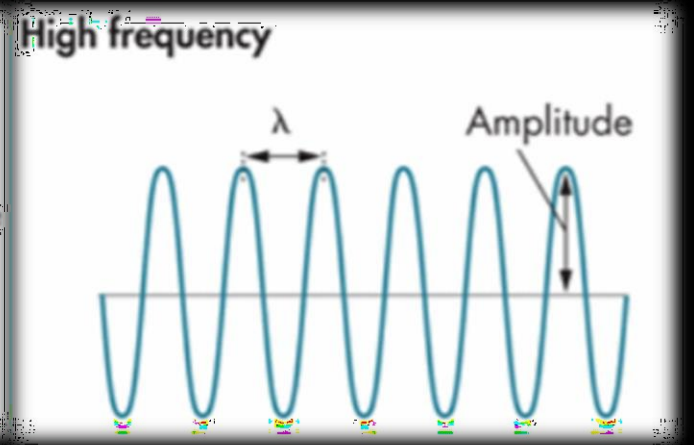
The **frequency (ν)**, is the number of wave cycles to pass a given point per unit of time. **Its unit is the s^{-1} or Hertz (Hz)**



The **frequency (ν)** and the **wavelength (λ)** of light are inversely proportional to each other

As the wavelength increases, the frequency decreases.



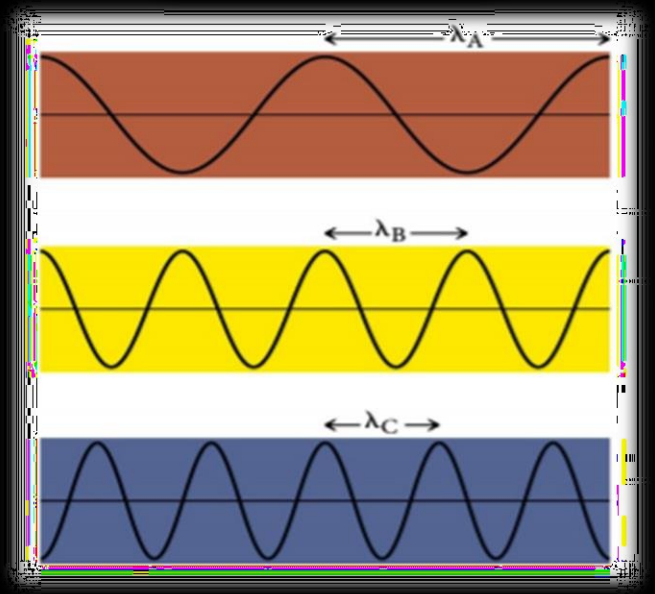


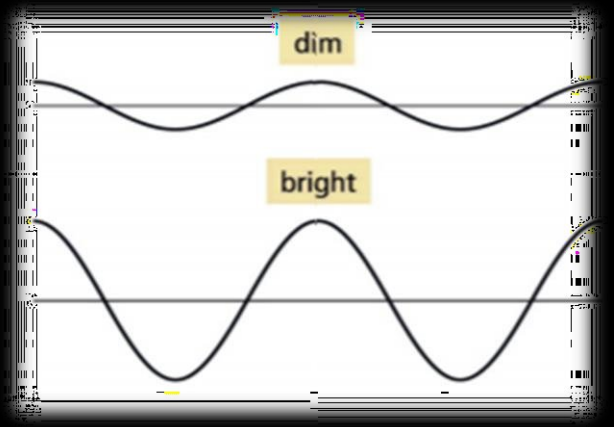
Nature of Light

$$c = \lambda v$$

$$v = \frac{c}{\lambda} = \frac{1}{T}$$

$$\ddot{v} = \frac{1}{\lambda} = \frac{v}{c}$$





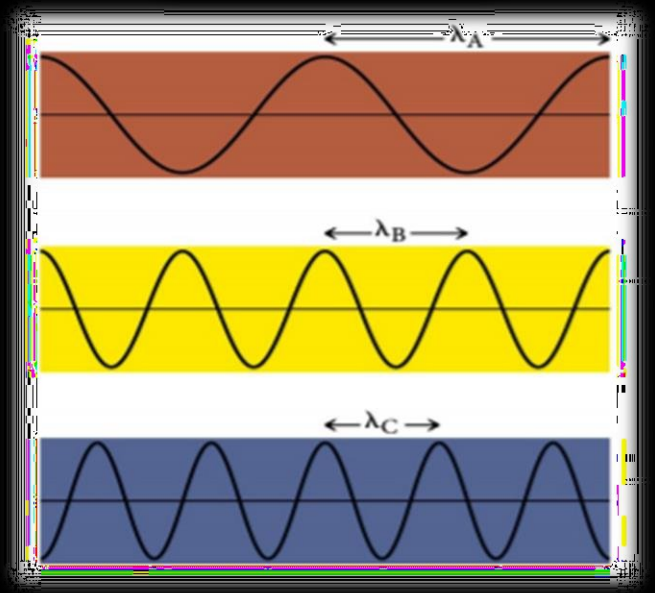
The colour of light is determined by **its wavelength (or frequency).**
The intensity is determined by **the amplitude.**

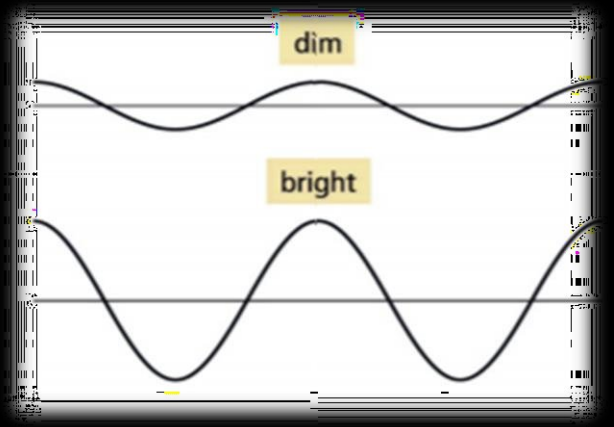
Nature of Light

$$c = \lambda\nu$$

$$\nu = \frac{c}{\lambda} = \frac{1}{T}$$

$$\ddot{\upsilon} = \frac{1}{\lambda} = \frac{\nu}{c}$$



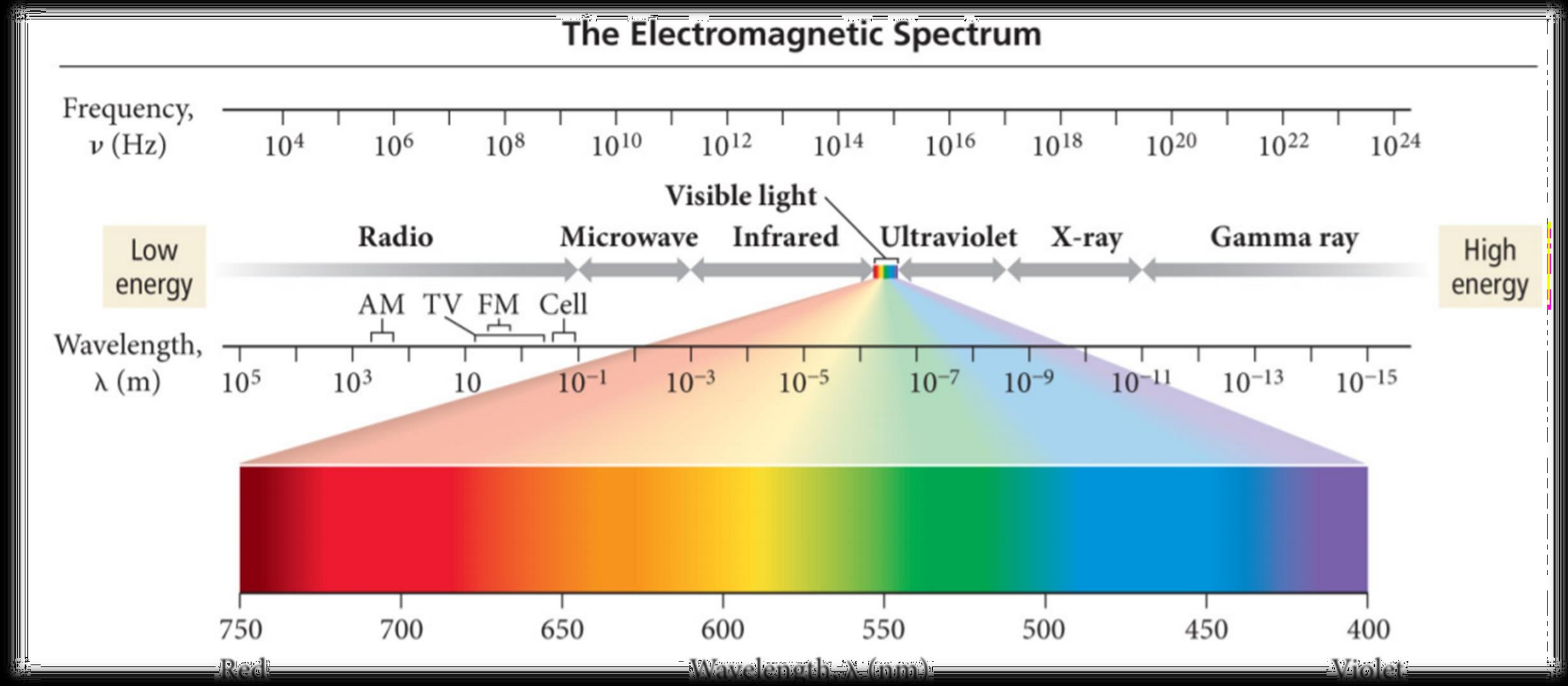


The wavenumber ü, is used in spectroscopy and most chemistry field, it is defined as the number of wavelengths per unit of distance, usually (cm^{-1})

A spectroscopic wavenumber can be converted into energy per photon

$$E = hc\ddot{\upsilon}$$

Nature of Light



Nature of Light
Exercises

$h = 6.626 \times 10^{-34}\ m^2\ kg\ /\ s$
$c = 299\ 792\ 458\ m\ /\ s$
$1\ J = 6.241509 \times 10^{+18}\ eV$

Frequency = 5.10^{14} Hz
Wavelength = ?

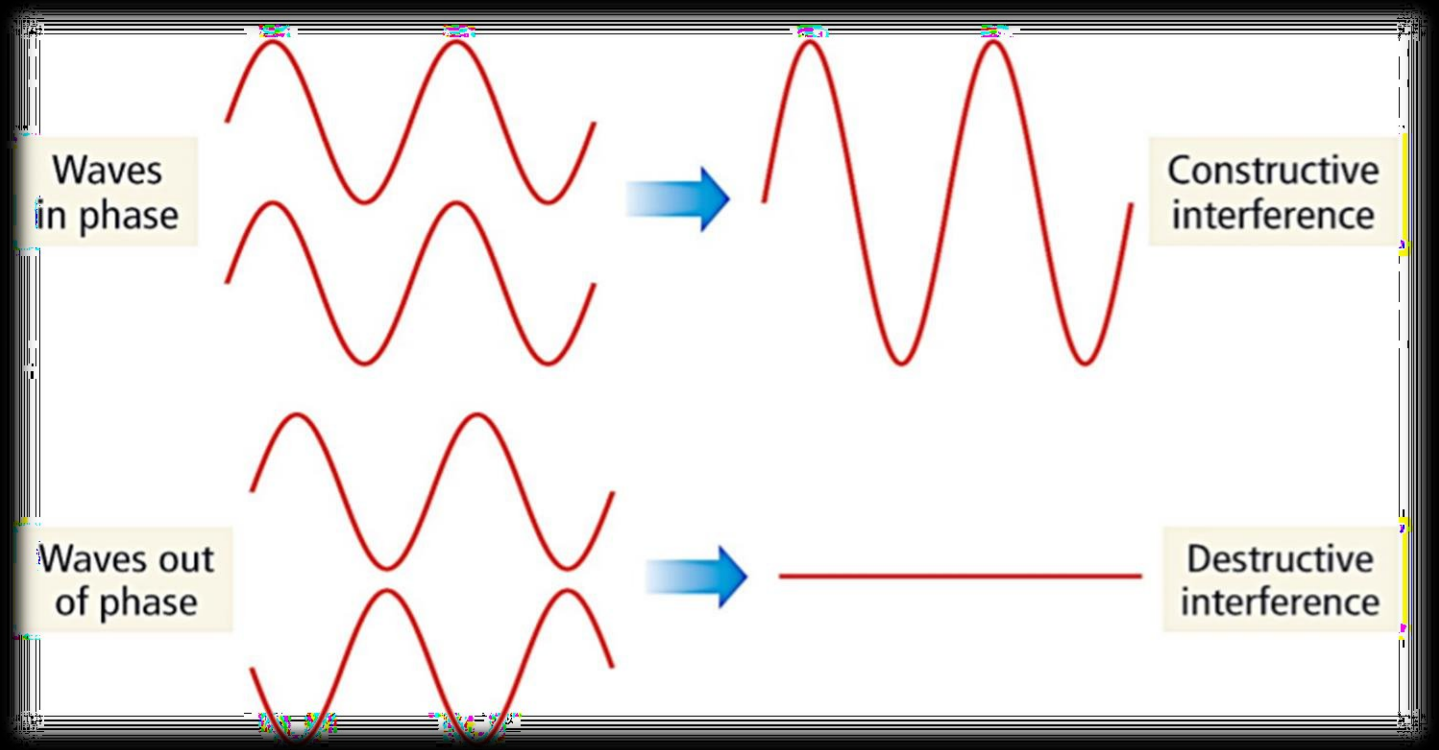
Energy (J) = ?

Energy (eV) = ?

Nature of Light

Interference and diffraction

When two waves overlap, their amplitudes add. This superposition is called interference.

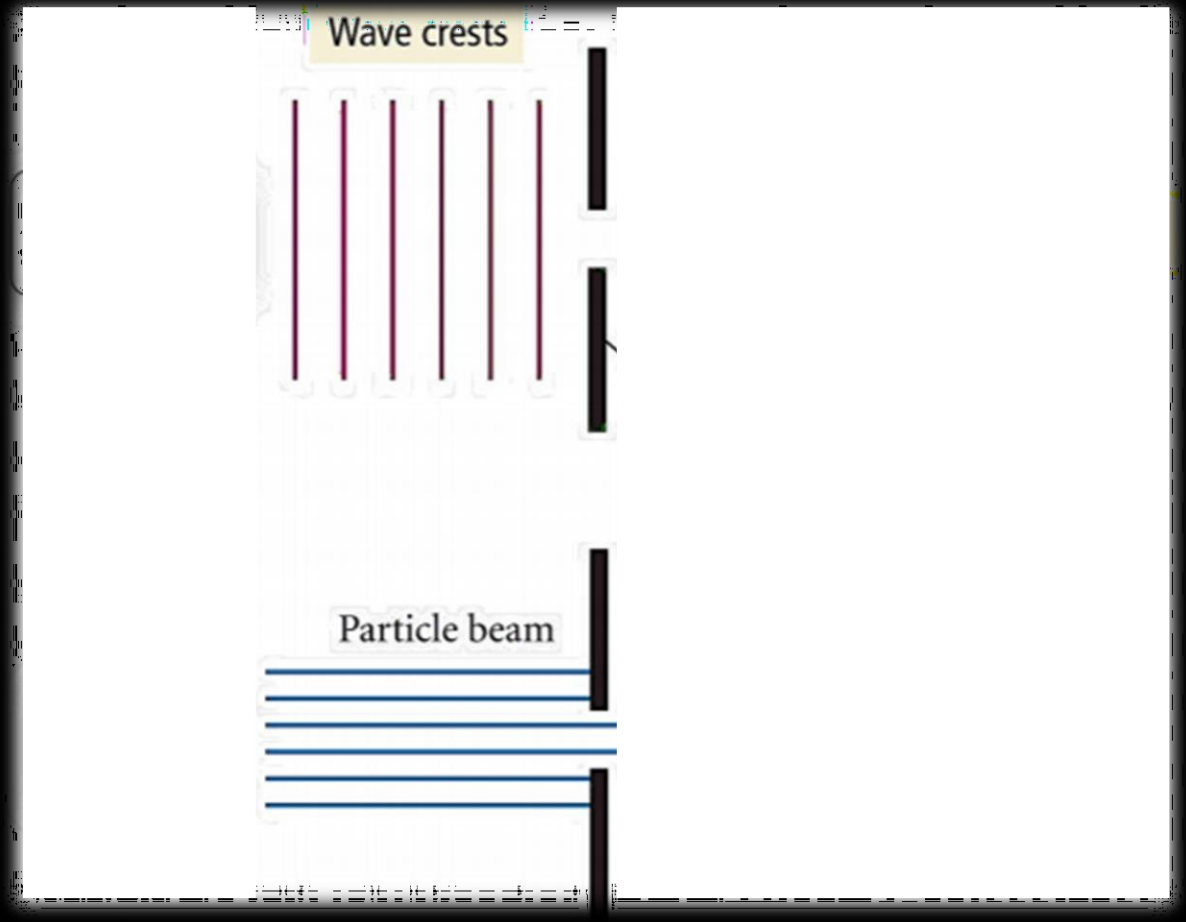


Nature of Light

Interference and diffraction

When traveling waves encounter an obstacle or opening in a barrier that is about the same size at the wavelength, they bend around it. This is called **diffraction.**

Traveling particles do not diffract

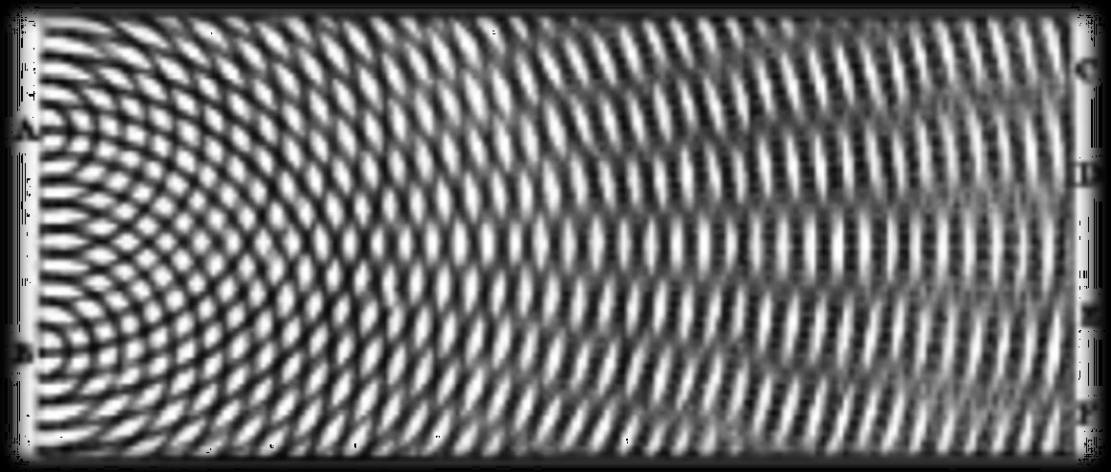


Nature of Light

Young's interference experiment (1802)

The diffraction of light through two slits separated by a distance comparable to the wavelength results in an interference pattern of the diffracted waves.

An interference pattern is a characterisic of all light waves.



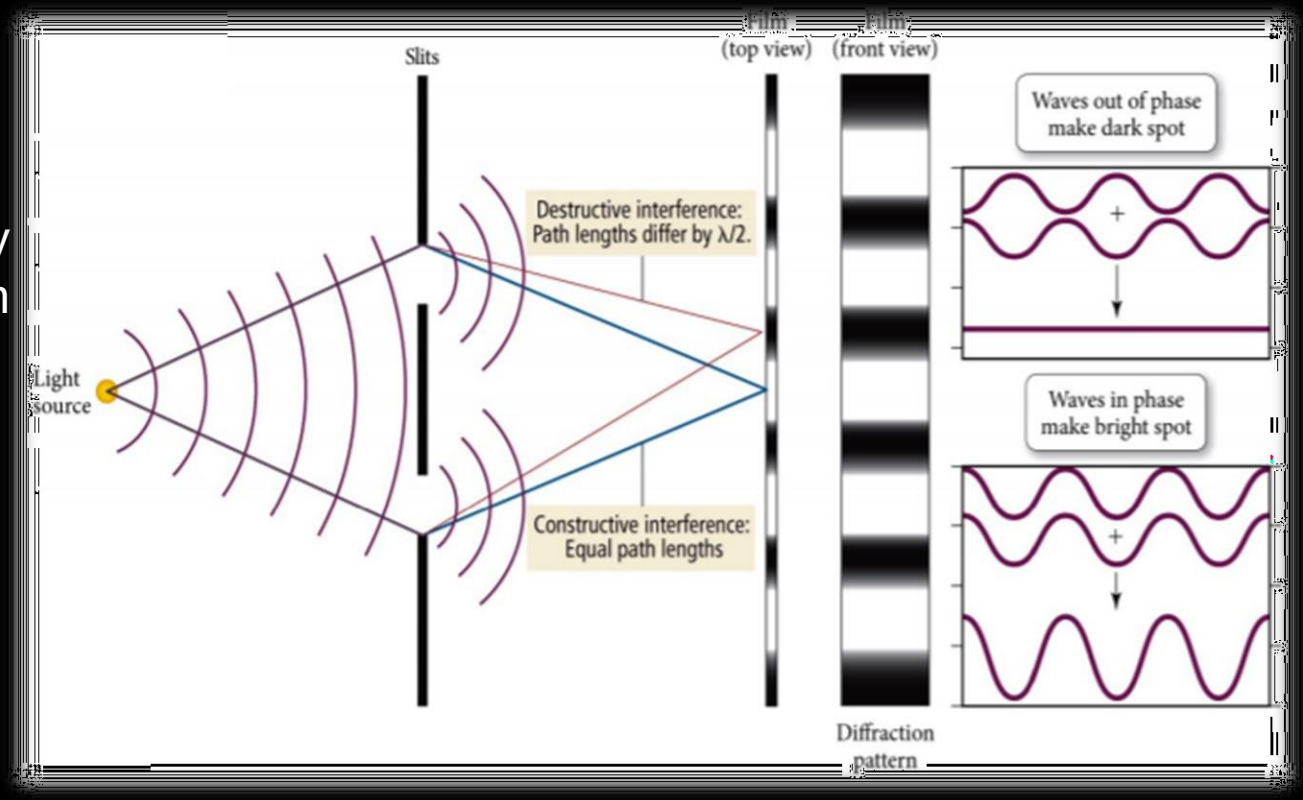
T. Young "Course of Lectures on Natural ¨Philisophy and the Mechanical Arts", 1807.

Nature of Light

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https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_all.html

The blackbody problem

Ultra violet catastrophe

A blackbody is a theoretical object that emits and absorbs radiation. When heated, the intensity and energy of its emitted radiation increase. (ex: Iron nail)



When an iron nail is heated with fire, it first stay the same color, but heat is radiating in the form of infrared radiation. (IR is invisible). When heated to a higher temperature, it will turn red (visible) because it is emitting higher energy waves.

For any blackbody, higher temperatures produce more energetic radiation of higher intensity.

The Planck Postulate

Quanta of Light



Planck postulated from study of radiation from blackbodies that the radiating atoms can **only radiate energy in discrete amounts** or that atoms exist only in discrete states called quantum states.

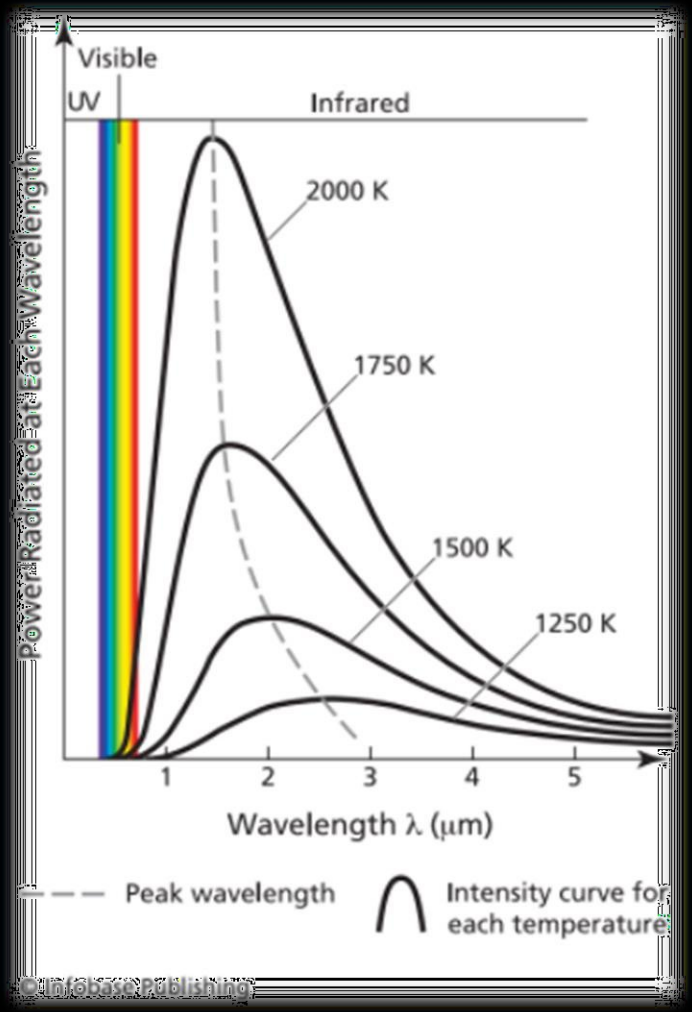
This was the birth of **quantum physics** in 1900.

Energy, like matter is dicontinuous

Planck **quantifies** the energy of light Light can only have energies given by :

$$E = h\nu$$

With $h = 6.626 \times 10^{-34}$ J.s



The Photoelectric Effect

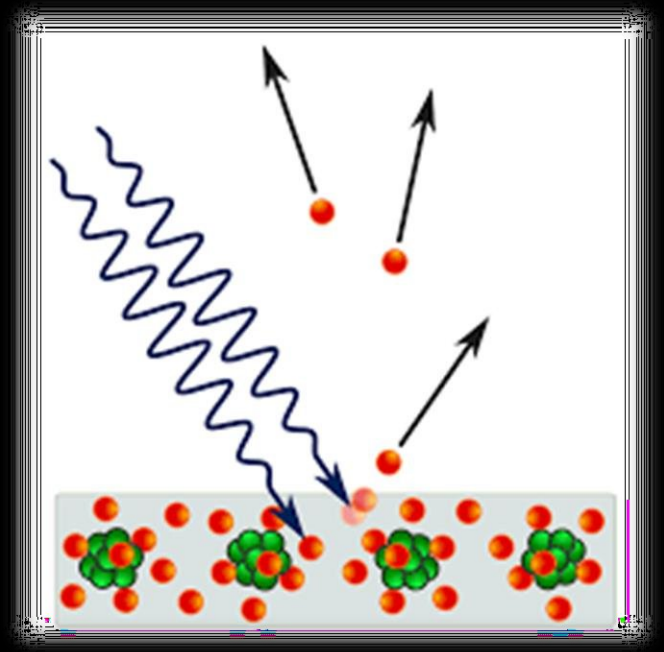
Light a particle

An other mystery for the period was discovered and named **the photoelectric effect.**

When light of a single color (or one wavelength) is shined on certain metals electrons are knocked out of the metal.

This experiment went against the Maxwell's theory that light is an electromagnetic wave

The energy of the ejected electrons should be proportional to the intensity of the light and independant of the frequency.



https://phet.colorado.edu/sims/cheerpj/photoelectric/latest/photoelectric.html?simulation=photoelectric

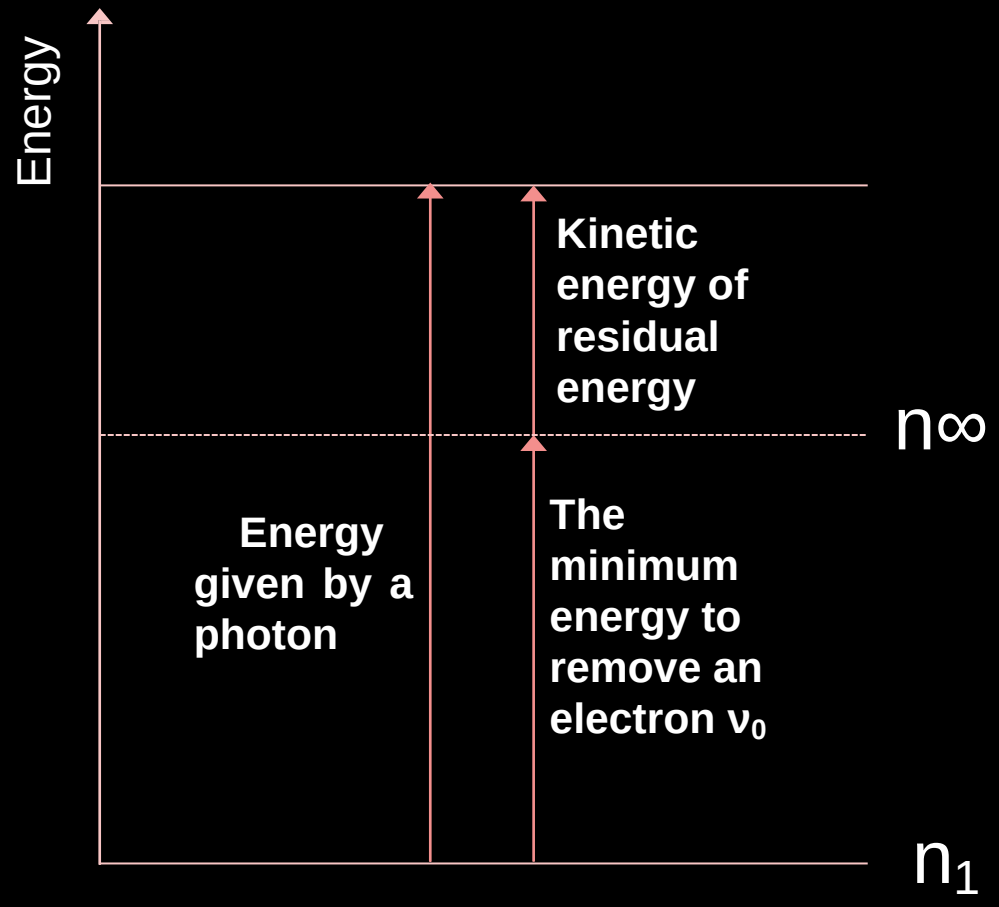
The Photoelectric Effect

Light a particle

In reality the energy of the ejected electrons is **independent of the intensity** and **depends directly on the frequency,** the number of electrons is proportional to intensity; their kinetic energy is not.

A minimum energy is needed before an electron can be emitted, called **the threshold frequency (ν_0)**

One photon at ν_0 gives the electron just enough energy for it to escape the metal, called Binding Energy



The Excess energy = Kinetic energy of the ejected electron

Kinetic Energy = $E_{photon} - E_{binding}$ $E_K = h\nu - h\nu_0$

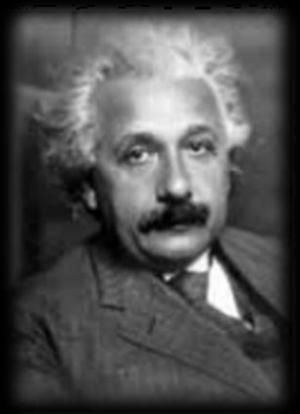
The Photoelectric Effect

Exercise

Kinetic Energy = E_{photon} – $E_{binding}$ $E_K = hv - hv_0$

If $E_K = 2.2 \cdot 10^{-19}$ J
what will be the velocity of the electron ?

Nature of light



In 1924 Einstein wrote: “There are therefore now two theories of light, both indispensable, and… without any logical connection.”

The Duality of light

- **Wave (Diffraction and Interference)**
- **Particle (Photoelectric effect)**

- Electromagnetic radiation characterized by **wavelength (m)** and **frequency (s^{-1})**
- Photon energy is quantized and its value is **E = hv**

J J.s s^{-1}

Atomic Spectrum of Hydrogen

Atomic Spectroscopy: Study of the interaction of atoms with electromagnetic radiation

White light: Continuous spectrum (continuous distribution of all wavelengths) The

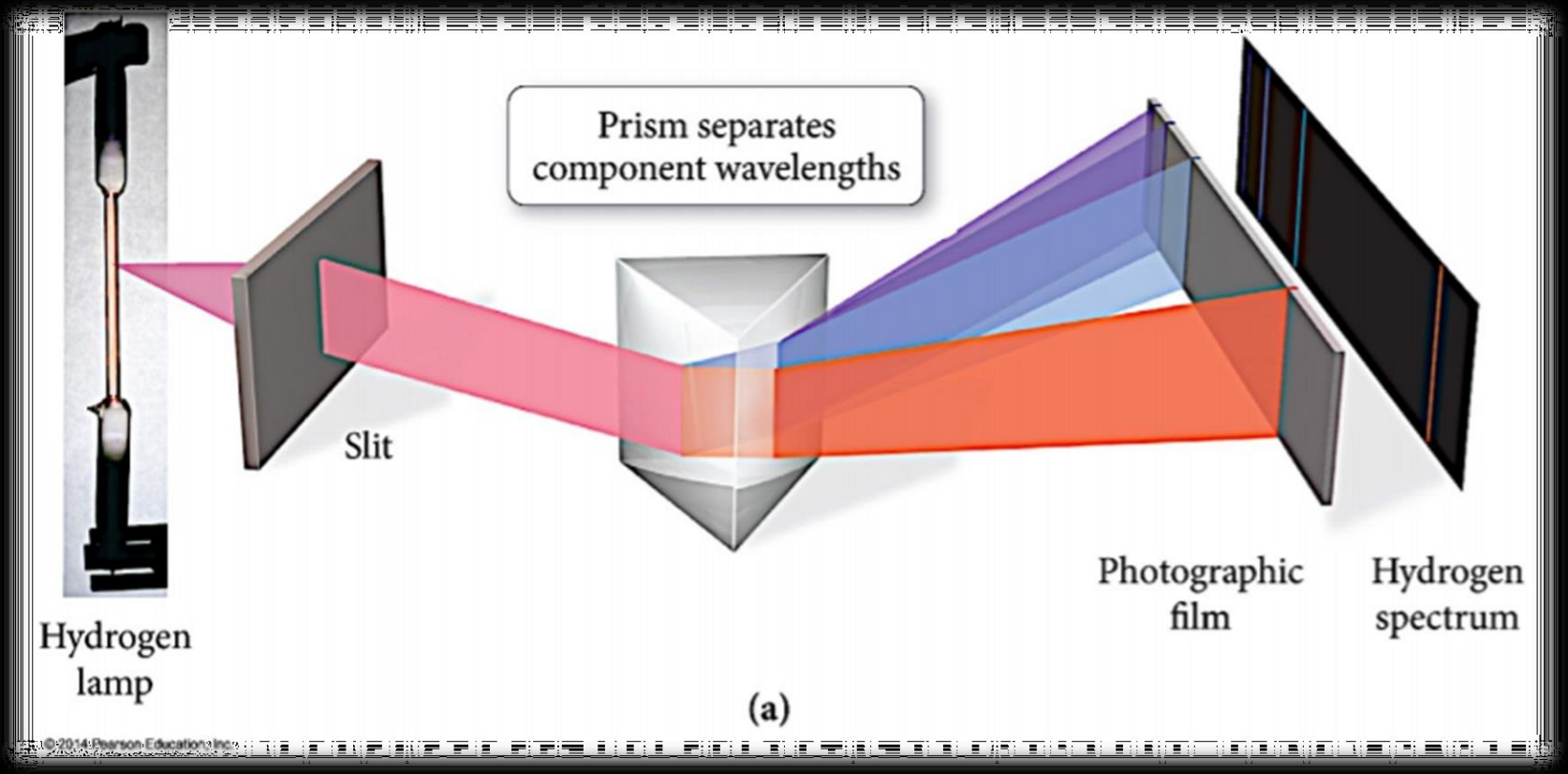
White light can be decomposed into its elements by a prism



White Light Spectrum



Emission Atomic Spectrum of Hydrogen



Only certain wavelength are emitted: spectrum appears as **sharp lines**

Emission Atomic Spectrum



Hydrogen Spectrum



Helium spectrum

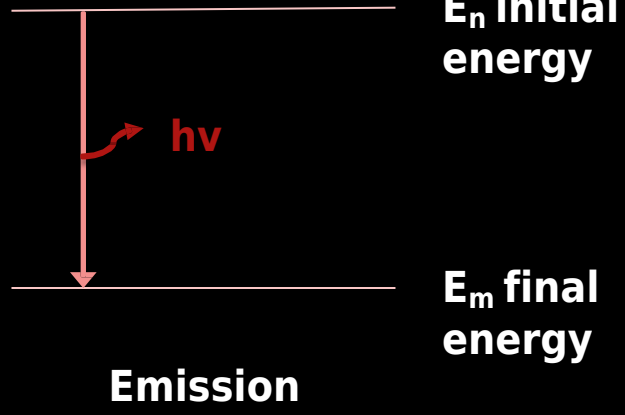


Barium Spectrum

Atomic Spectrum of Hydrogen

Electronic Energy of the atom is **quantized: not all values are allowed**

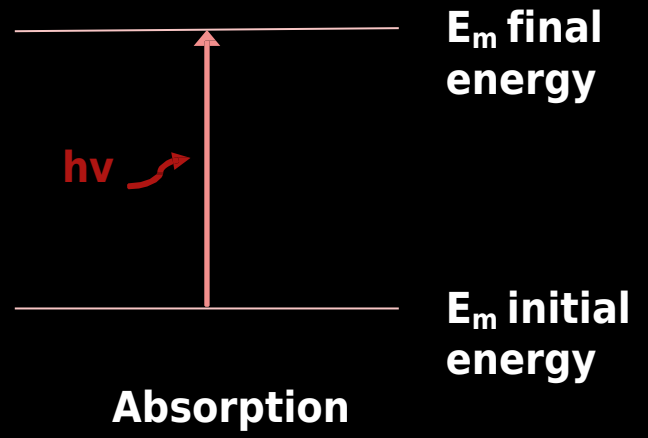
The lines observed in atomic spectra correspond to **electronic transitions** between **energy levels**



Emission

Emitted Photon

$$E_{photon} = E_n - E_m = h\nu$$



Absorption

Absorbed Photon

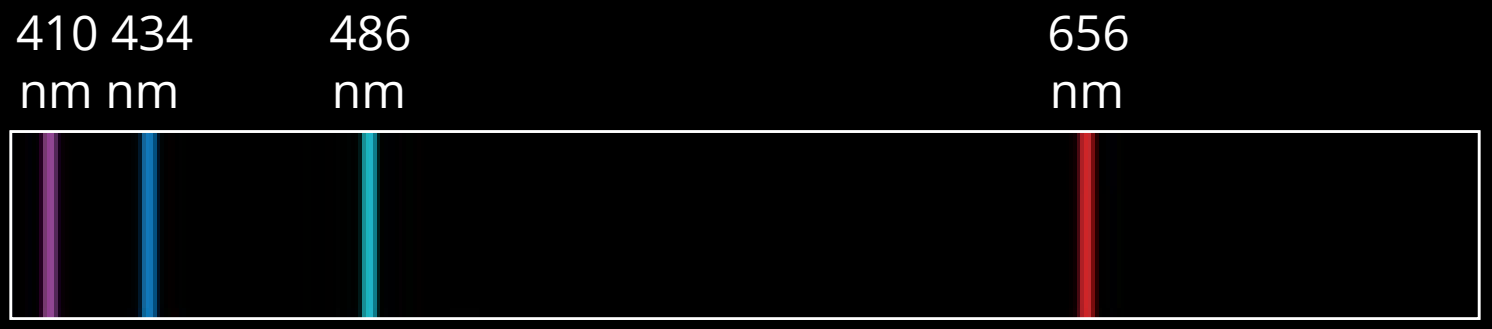
$$E_{photon} = E_n - E_m = h\nu$$

Empirical Rydberg Law

Johannes Rydberg analysed the spectrum of hydrogen and found that it could be described with an equation that involved an inverse square of integers.

$$\frac{1}{\lambda} = R_H \cdot \left(\frac{1}{m^2} - \frac{1}{n^2} \right) \quad m<n$$

Rydberg constant $R_H = 1.097.10^7\ m^{-1}$



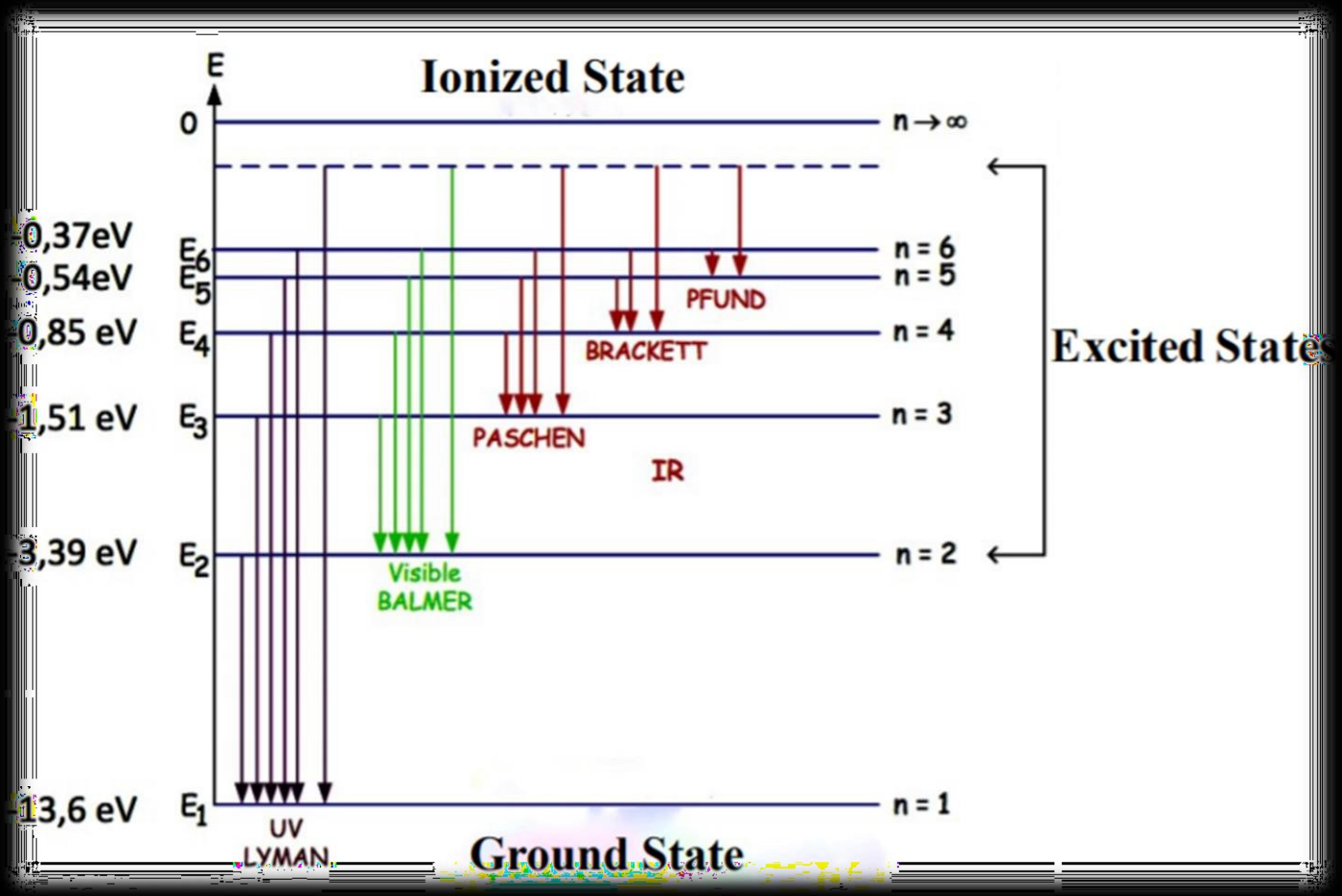
Empirical Rydberg Law

For Hydrogen atom, series have the name of the scientist whom studied it

Series: group of lines corresponding to n_1
With $n_x = n_1+1, n_1+2, n_1+3, n_1+4,\ldots$

Chronology

Lyman 1906-1914
Balmer 1885
Paschen 1908
Brackett 1922
Pfund 1924



Empirical Rydberg-Balmer Law

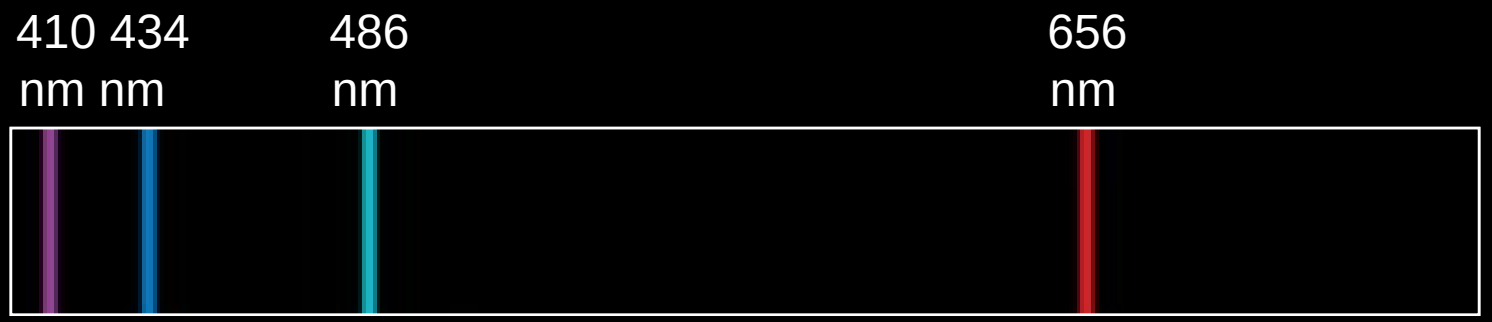
The visible

In the visible domain, the wavelengths of the hydrogen atom lines are given by

$$\frac{1}{\lambda} = R_H \cdot \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

$n_1 = 2$
$n > 2$

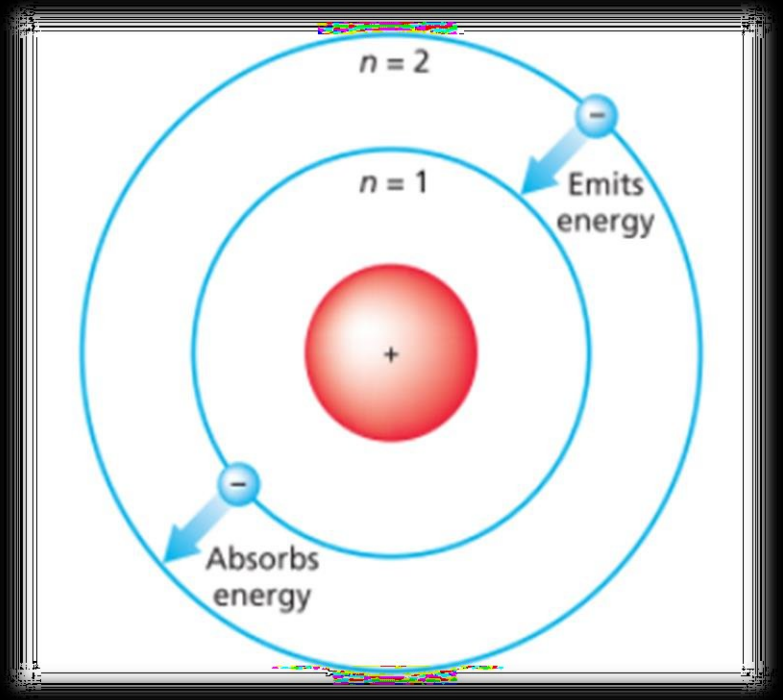
Rydberg constant $R_H = 1.097.10^7\ m^{-1}$



Niels Bohr The Quantic Model

Bohr student of Rutherford was following the work of Planck and Einstein.
And try to apply the quantification observed in blackbody and photoelectric effect.
The **energies of electrons in an atom must not be continuous**, and **must only take specific value?**

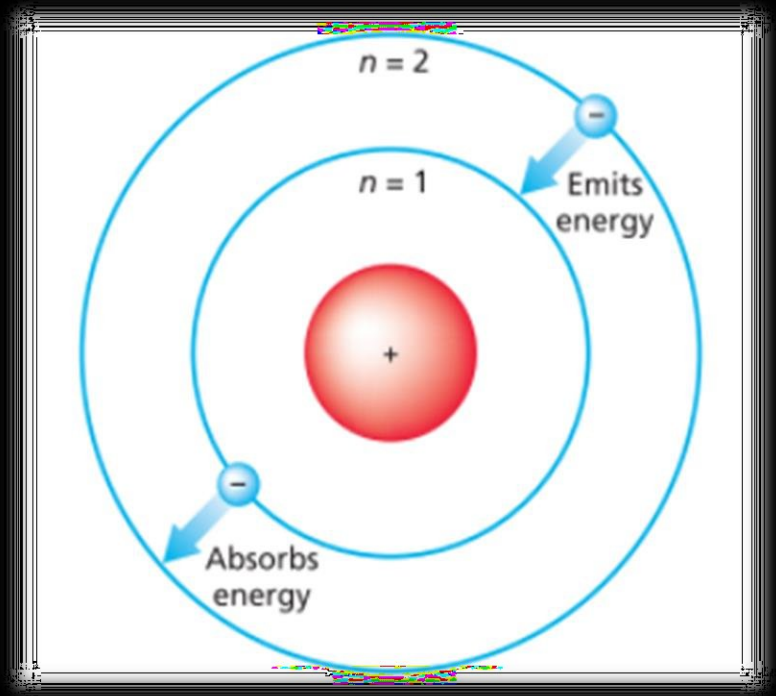
Bohr proposed that an electron is found only in specific circular paths, or orbits, around the nucleus



“Solar System” model

Bohr Atomic Model

- The electron moves in **circular orbits** around the nucleus.
- The **stationary states** are the **allowed orbits where the electron can move**. In a stationary state the **energy of an electron is constant** and no energy is emitted.
- By **ejecting a photon (emitting energy**), an electron can move from a **higher to a lower energy level**. Conversely, by **absorbing energy**, an electron can jump from a **lower energy level to a higher one.**
- There are **no intermediate energy levels**. The electron is either in one level or the other and moves instantaneously between the two.



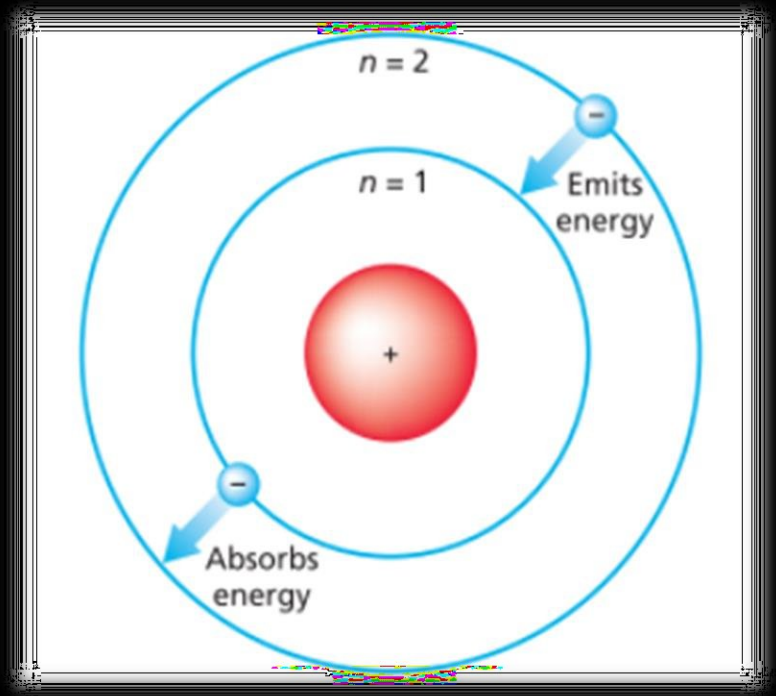
“Solar System” model

Bohr Atomic Model

- Radii of the stationary states for the Hydrogen atom
- $\mathbf{R_n = a_0 \,.\, n^2}$ with a_0 = Bohr radius = 52.9 pm
- Energy levels of Hydrogen atom

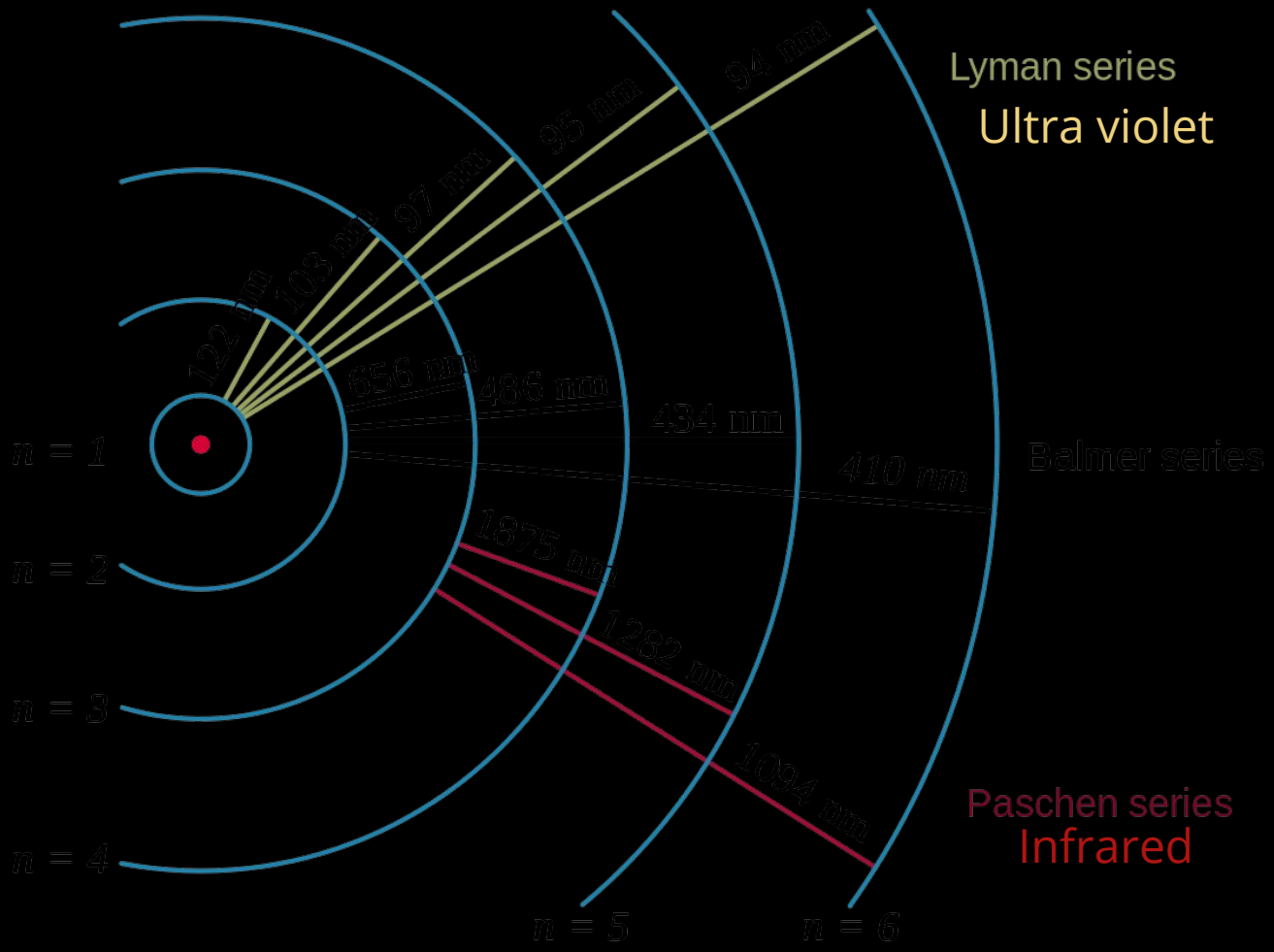
$$E_n = -\frac{13.6\ eV}{a\,J n^2} = -\frac{2.1799}{n^2}$$

with n = principal quantum number



“Solar System” model

Bohr Atomic Model



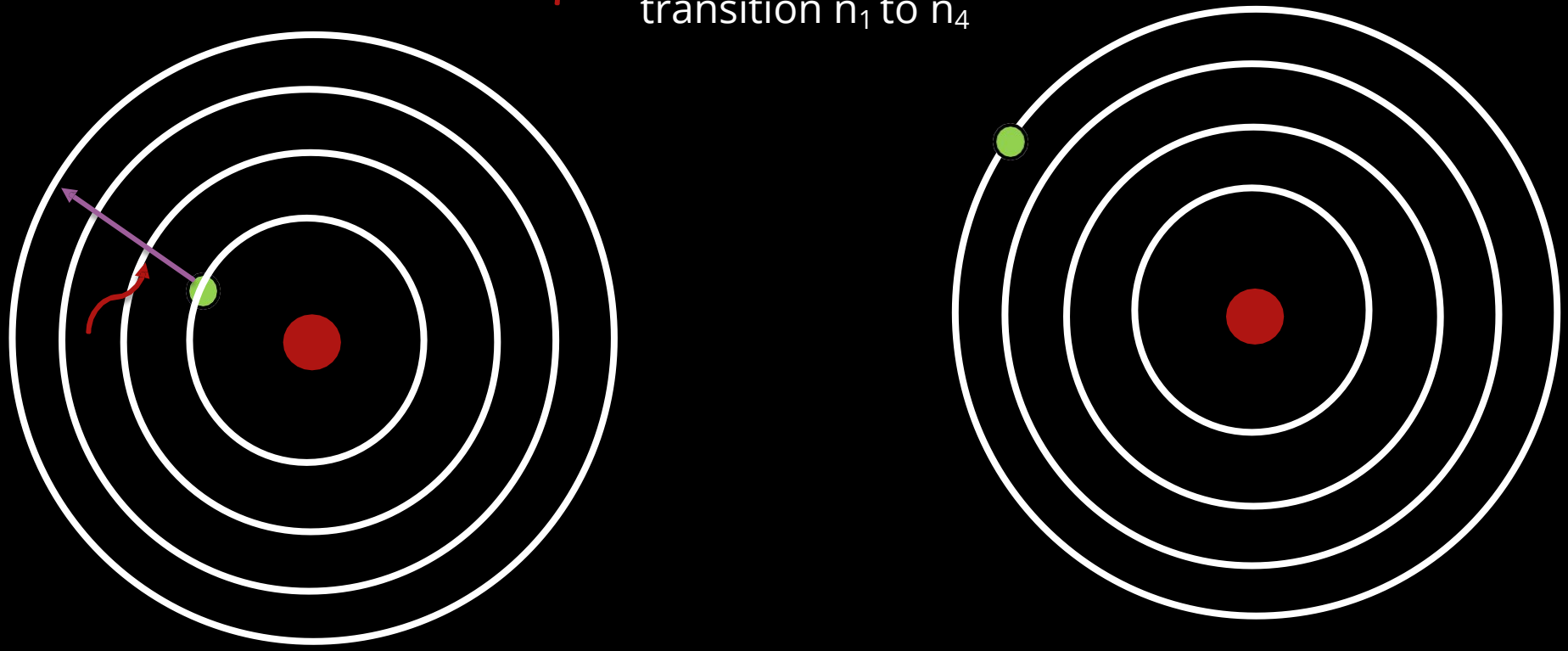
Bohr atomic model can explain the Rydberg Law

$$\frac{1}{\lambda} = R_H \cdot \left(\frac{1}{m^2} - \frac{1}{n^2}\right)$$

Bohr Atomic Model

The absorption

Energy absorption
Exactly equal to the transition n_1 to n_4

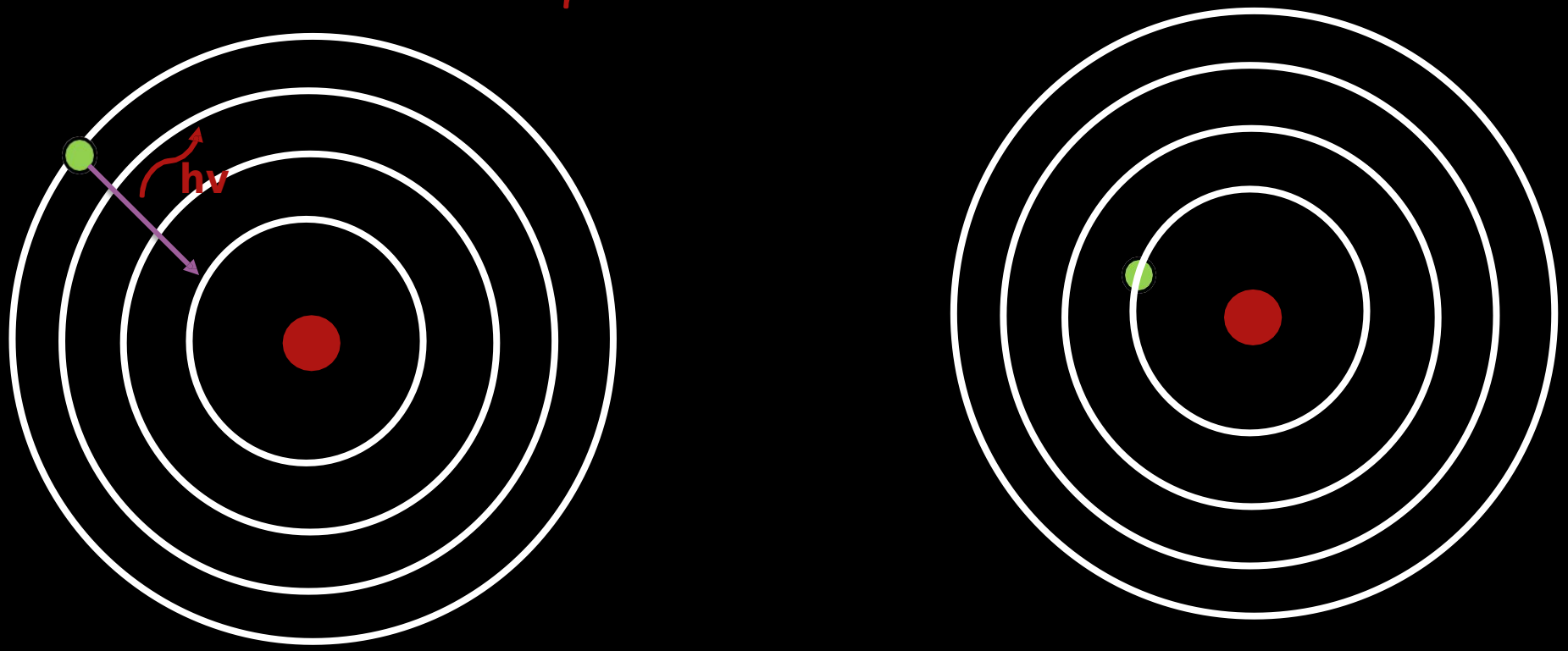


The lowest energy level is called the **ground state.**
When an atom gains energy, it is in an **excited state.**

Bohr Atomic Model

The emission

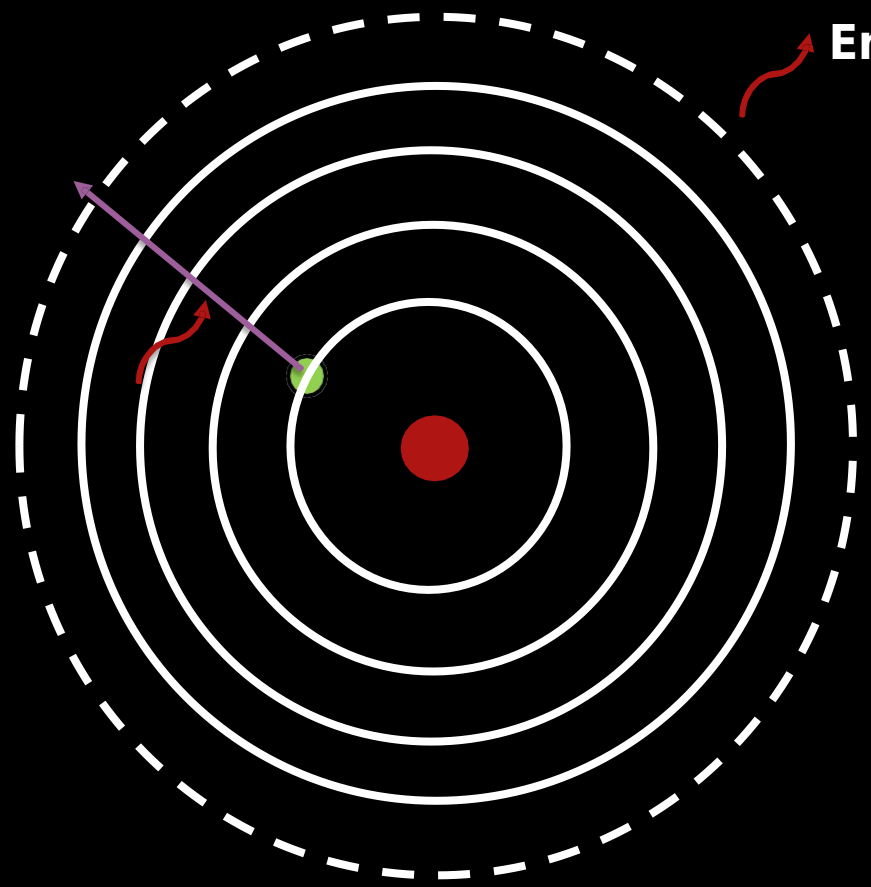
Energy emission (light, uv)



When the electron falls back to its ground state it release energy (hv)

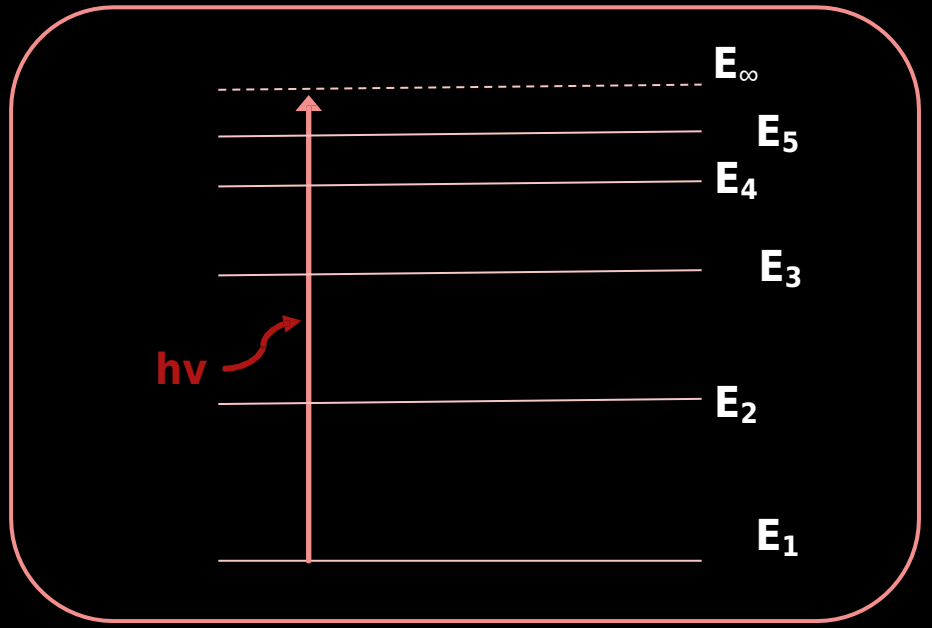
Bohr Atomic Model

What happen if you give " a lot of energy" to an electron ?



The atom is ionized

$$E_i(H) = E_\infty - E_1 = +\,13.6\ eV$$



Ionization Energy: minimal energy required to remove one electron from the atom in its ground state.

Hydrogen-like atoms

Hydrogen like atoms:
Is an **ion** that mimic the hydrogen (only one electron) with atomic number Z and nuclear charge +Ze

Ex : He^{+}, Li^{2+}, Be^{3+}, C^{5+}.

Z = atomic number of the hydrogenoid ion

$$\frac{1}{\lambda} = R_H \cdot \left(\frac{1}{m^2} - \frac{1}{n^2}\right) \cdot Z^2$$

m<n

Rydberg constant $R_H = 1.097.10^7\ m^{-1}$

Rydberg equation

$$E_n = -\frac{13.6\ eV}{n^2} \cdot Z^2$$

Energy of Hydrogen-like atom

$$r_n = \frac{a_0}{Z} \cdot n^2$$

Radius of stationary orbits

Hydrogen-like atoms

$$\frac{1}{\lambda} = R_H \cdot \left(\frac{1}{m^2} - \frac{1}{n^2}\right) \cdot Z^2$$

Rydberg equation

m<n

Rydberg constant $R_H = 1.097.10^7\ m^{-1}$

$$E_n = -\frac{13.6\ eV}{n^2} \cdot Z^2$$

Energy of Hydrogen-like atom

Calculate the Energy of an Hydrogen electron moving from n_3 to n_2:

Bohr Atomic Model

Summary

1. In the shell model of Bohr, the electron is in an orbit.
2. The radius of these orbits is quantized and it is proportional to n2 (shell number or principal quantum number)
3. The energy of the hydrogen atom is negative (convention: free electron has zero energy)
4. The energy of the hydrogen atom is quantized and it depends on the shell n (where electron is)
5. The more the shell is closer to nucleus, the more its energy is negative

Bohr Atomic Model

Summary

6. Shells corresponding to the highest n are the most energetic

7. As n increases, the radius of the shell increases, and so the electron is less attracted by the nucleus.
If n $\longrightarrow \infty$ the electron has been ejected: atom is ionized

6. As n increases, the energetic levels get closer

7. The lowest energy level corresponds to n=1. It is the ground state. All other are excited states

8. The Ionization energy is the minimum energy needed to eject an electron from an atom. It depends on the atom state (ground or excited state)

Bohr Atomic Model

Limits

1. Not valid for multi-electron atoms
2. Does not predict fine structure of atomic spectral lines
3. It does not predict variations in the intensity of spectral lines
4. Wrong value of angular momentum for electron in atom
5. It violates Heisenberg uncertainty principle
6. Electron, due to movement, should lose energy (as light emission) and collapse with the nucleus

Electron particle and wave



Following the dual nature of light, Louis de Broglie came up with a bolder idea :
If light can be a particle and a wave, what about electrons ?

His hypothesis was even bigger, he said that **all matter,** from electron to basketballs, has both wave and particle characteristics.

His equation showed that in larger bodies (visible) the wave character was negligible.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

λ = Broglie wavelength (m)
h = Planck constant (J· s)
p = quantity of motion: momentum (kg· m· s^{-1})
m = electron mass (kg)
v = speed of the electron (m· s^{-1})

The solution to the wave equation led to an atom governed by probabilities.

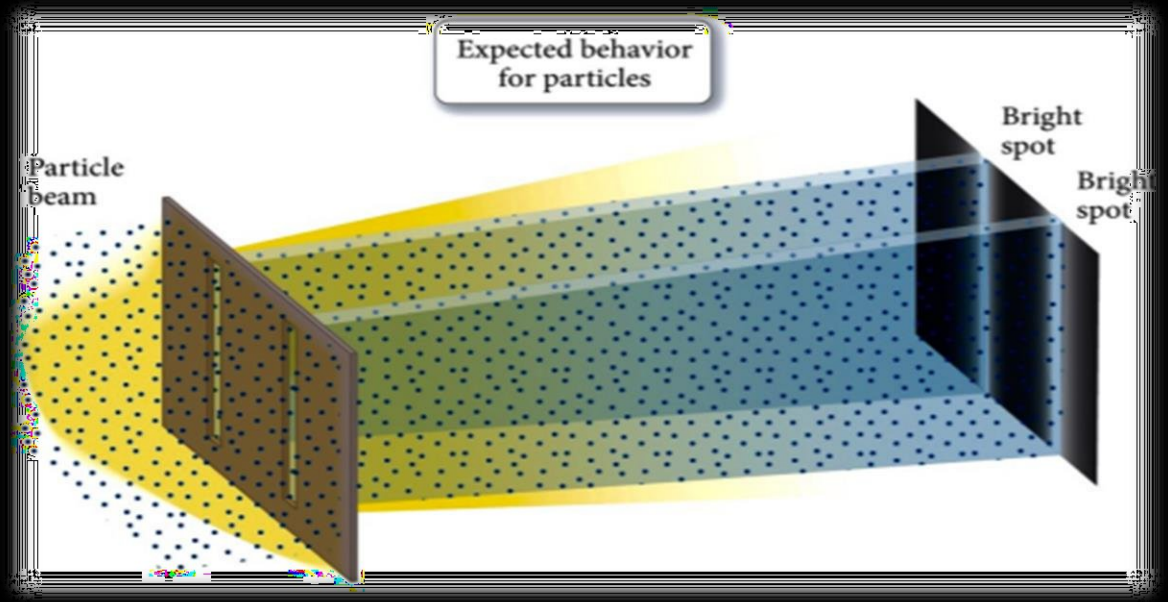
Electron particle and wave

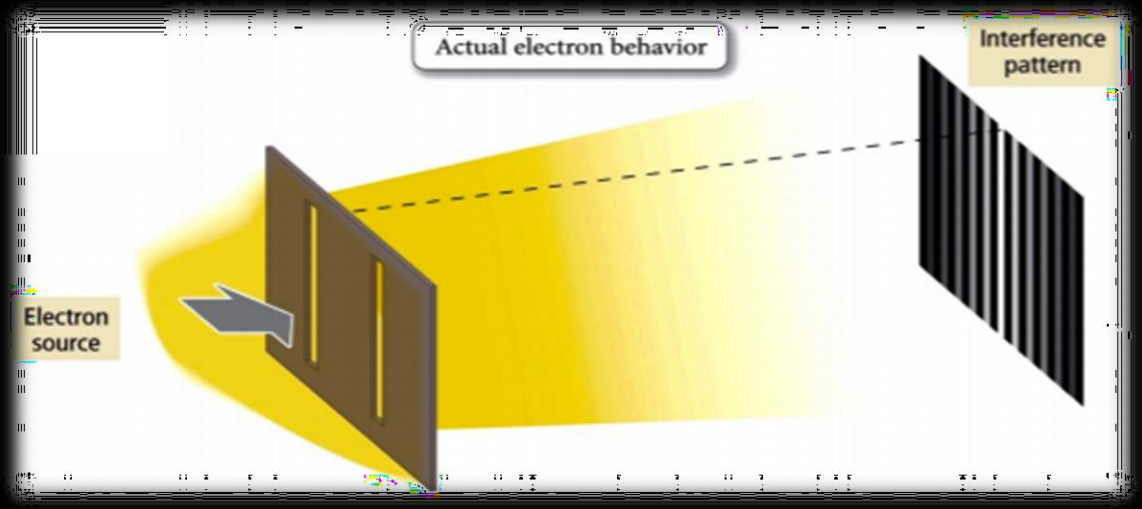
The experiment 1927

Clinton Davisson and his junior partner Lester Germer in the United States and George Thomson in Great Britain made the discovery. For their work, Davisson and Thomson were awarded a Nobel Prize in1937.

Fun Fact :

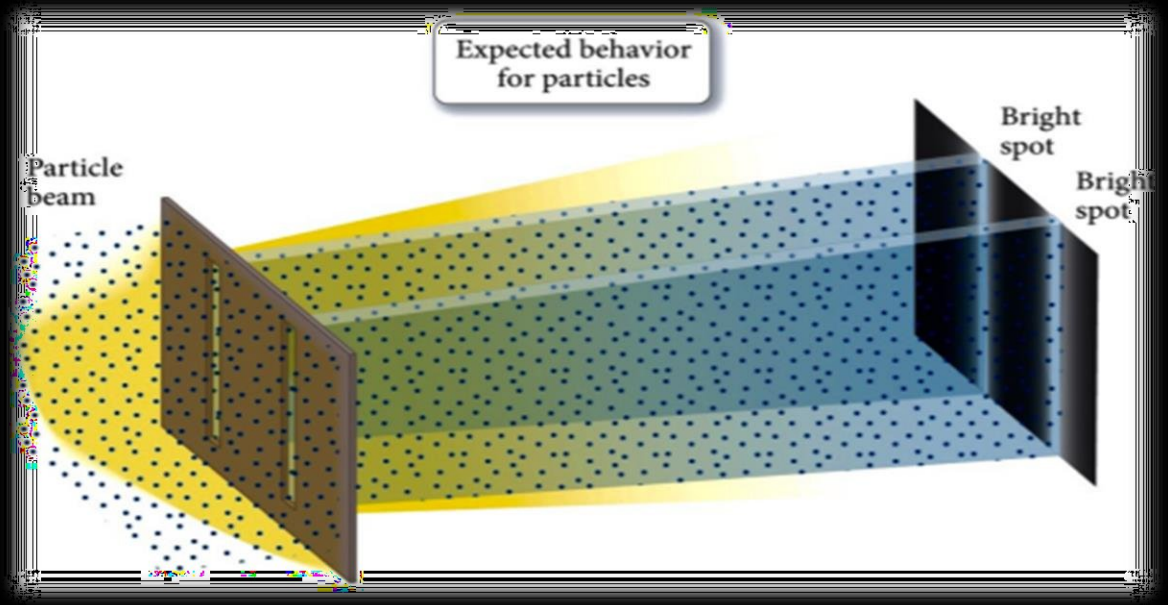
George Thomson was J.J.'s son. So J.J. won his Nobel for proving that electrons were particles. His son won for proving they were waves.

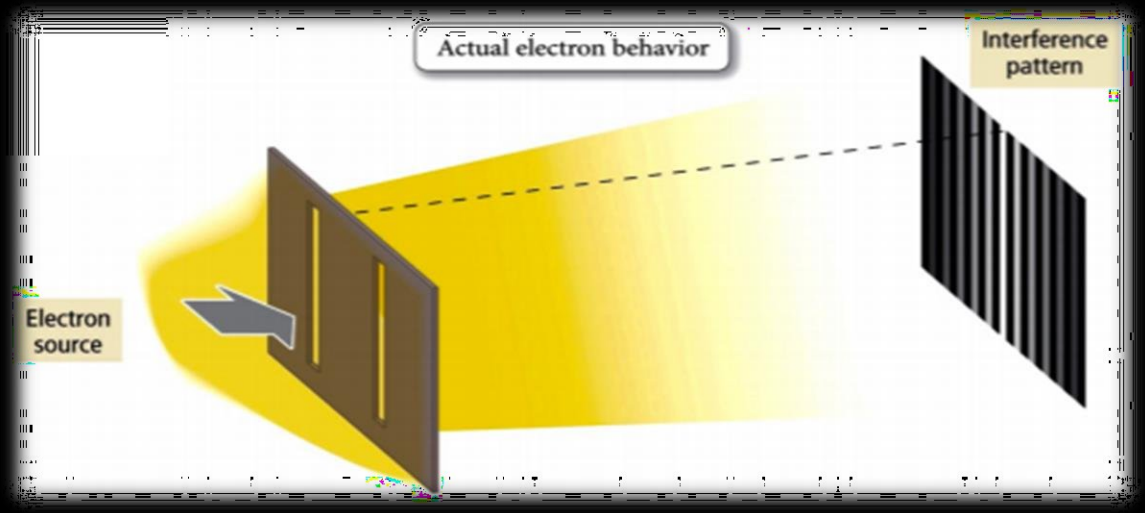




Electron particle and wave

The experiment 1927





Nickel bombarded by electrons generates same pattern as the one predicted for X-ray (electromagnetic wave)

Uncertainty Principle

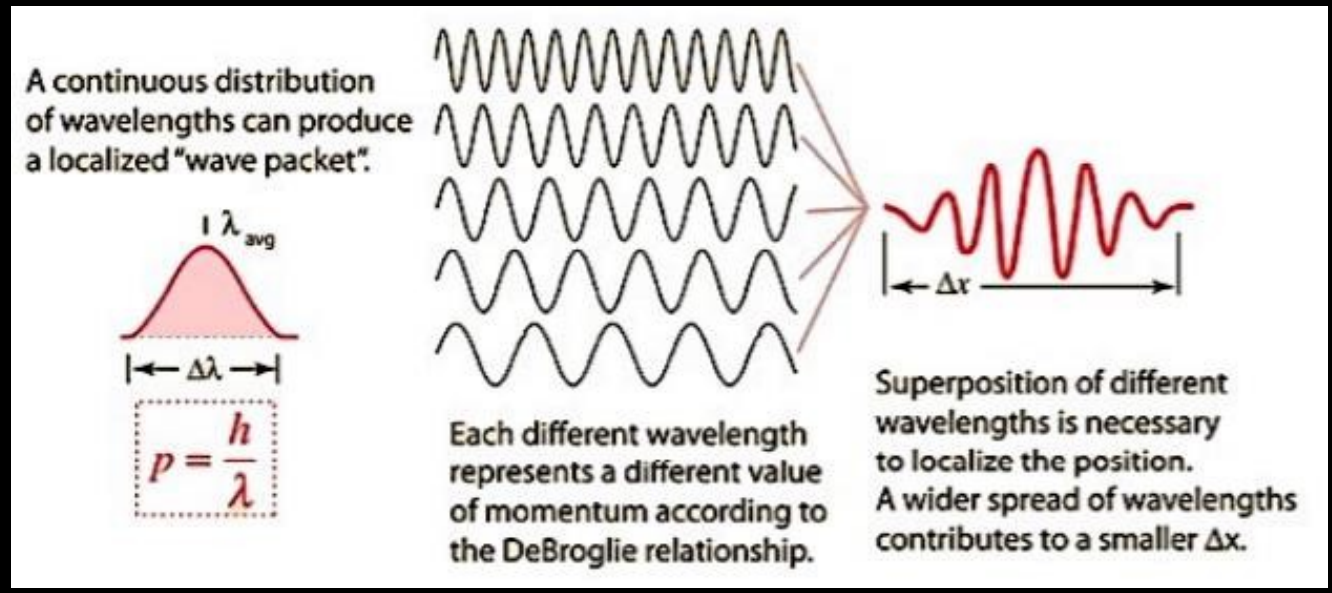
Heisenberg



He showed that uncertainty was a fact of the atomic world. According to Heisenberg, one **could not measure** precisely both **the position and the momentum** of a particle. Although his famous uncertainty principle can be stated in many ways, the most common mathematical expression is

$$\Delta p \times \Delta x \geq h/4\pi$$

Δp = uncertainty in the quantity of motion
Δx = uncertainty in the position
[p = quantity of motion: momentum ($kg \cdot m \cdot s^{-1}$)]



Uncertainty Principle

Heisenberg

Example 1
Suppose that the position of Khabib Nurmagomedov (mass 70 kg) is known within ±1 m.
What is the uncertainty in his velocity?

Uncertainty Principle

Heisenberg

Example 1
Suppose that the position of Khabib Nurmagomedov (mass 70 kg) is known within ±1 m.
What is the uncertainty in his velocity?

$$\Delta x = 1\ m$$

$$\Delta v \geq \frac{h}{(4\pi \cdot m \cdot \Delta x)}$$

$$\Delta v \geq \frac{6{,}632 \cdot 10^{-34}}{(4\pi \cdot 70 \cdot 1)}$$

$$\Delta v \geq 7{,}52 \cdot 10^{-37}\ m \cdot s^{-1}$$

Velocity uncertainty is negligible for a macroscopic object

Uncertainty Principle

Heisenberg

Example 2
Suppose that the Bohr's orbit radius a_0 = 52.9 pm is known within 1%.
What is the uncertainty in the velocity (Δv) of the electron with mass $9.1 \cdot 10^{-31}$ kg?

Uncertainty Principle

Heisenberg

Example 2
Suppose that the Bohr's orbit radius a_0 = 52.9 pm is known within 1%.
What is the uncertainty in the velocity (Δv) of the electron with mass 9.1·10-31 kg?

$$\Delta x = 52{,}9 \cdot 10^{-14}\ m$$

$$\Delta v \geq \frac{h}{(4\pi \cdot m \cdot \Delta x)} \rightarrow \Delta v \geq 1{,}09 \cdot 10^{8}\ m \cdot s^{-1}$$

$$\Delta v \geq \frac{6{,}632 \cdot 10^{-34}}{(4\pi \cdot 9{,}1 \cdot 10^{-31} \cdot 52{,}9 \cdot 10^{-14})}$$

Velocity uncertainty is huge (1/3 of the speed of light) for an electron If we know the position of the electron, we do not know its velocity

Schrödinger Model



It is impossible to determine the exact position of an electron in an atom.

Current understanding of the atom is based on Quantum Mechanics

This model sees the electrons not as individual particles, but as behaving like a cloud - the electron can be “anywhere” in a certain energy level.
If electrons are waves, their position and motion in space must obey a wave equation

$$\hat{H}\psi = E\psi$$

H: Hamiltonian operator which acts on the function Ψ and transforms it
E: energy of the electron

This equation has a solution only for certain values of the energy (quantized energy)
Solving the Schrödinger equation means finding the set of pairs (Ψ_i, E_i) that satisfy this equation

Schrödinger Model

$$\hat{H}\psi = E\psi$$

H: Hamiltonian operator which acts on the function Ψ and transforms it
E : energy of the electron



The Ψ-wave function obeys the Schrödinger equation (the transposition of $E = E_p + E_K$ from classical mechanics to quantum mechanics)

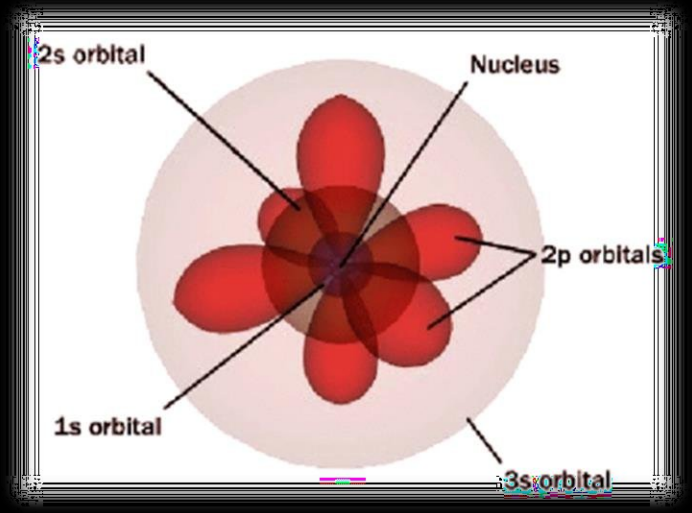
The wave function Ψ has no physical meaning.

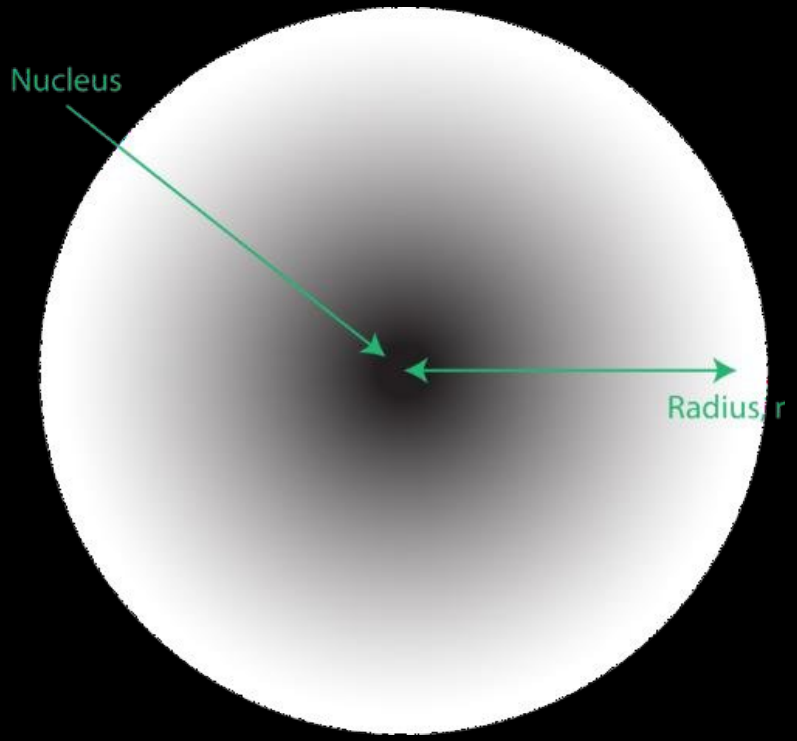
On the other hand, the value at one point of its square Ψ^2 (or of its square modulus $|\Psi^2|$)

determines the probability dP of finding the electron in a portion of the space denoted dv around this point

Atomic Quantum Model

- In the quantum model we do not attribute to the electrons a defined trajectory (as would be done if they were considered as particles)
- Each point of the diagram represents a possible position for the electron of the hydrogen atom in its ground state: it is the probability of presence
- Probability varies in function of the distance to the nucleus
- Probability can be different in different directions

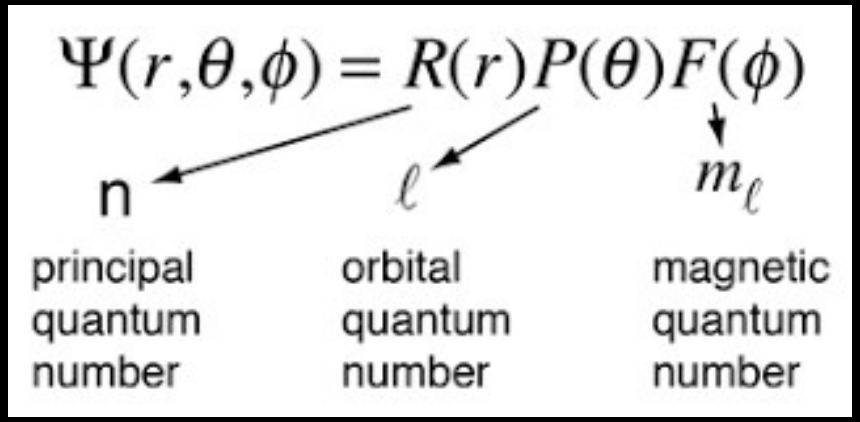




Solving Schrödinger equation
for Hydrogen atom

- The wavefunctions solution of the Schrödinger equation are the **Atomic Orbitals**
- The solution to these equations yields the **first three quantum numbers** and the limits on the values they can have.

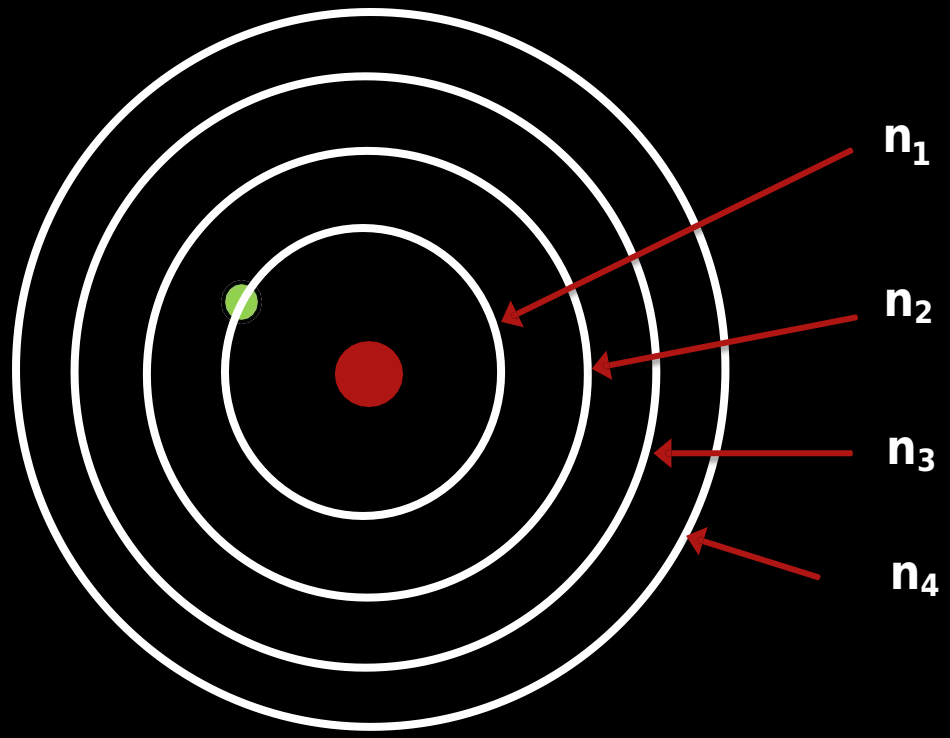
n, l, m are the three integers called **quantum numbers**



Quantum Numbers

n: The Principal Quantum Number

The principal quantum number for the **lowest energy shell is n = 1**, the next is **n = 2**, etc..
where **n is any positive whole number or integer**.
The higher the principal quantum number, the greater the energy of the electrons in that shell.



It is related to the average distance of the electrons from the nucleus.

Electrons with larger n values are more energetic and farther from the nucleus.

All electrons with the same value of n are in the same principal electronic shell

Quantum Numbers

l: The Orbital Angular Momentum Quantum Number

The second quantum number is called the **angular momentum quantum number**.
It is designated by the letter l. This quantum number governs the angular momentum of the electrons and **determines the shape of an orbital**, which indicates **where an electron is likely to be** in the atom.

l is an integers between 0 and n-1

ex: **If n = 3** There is 3 orbital shapes that could be present **l = 0, 1, 2 (s, p, d).**

Value of l (subshell)	Letter
0	s
1	p
2	d
3	f
4	g

All electrons with the same n and l values are in the same subshell or sublevel

Quantum Numbers

m_L: The Magnetic Quantum Number

The third solution to Schrödinger's equation produces the magnetic quantum number, usually designated as m_L

The magnetic quantum number specifies how the s, p, d, and f orbitals are oriented in space.

m_L is an integers values between + l and - l

ex: **If l = 1** There is 3 different orientation m_l **= - 1, 0, + 1** **(x, y, z).**

Value of l (subshell)	Value of m_l	Number of Orbitals
0	0	1
1	- 1 , 0 , + 1	3
2	- 2 , - 1 , 0 , + 1 , + 2	5
3	- 3 , - 2 , - 1 , 0 , + 1 , + 2 , + 3	7
4	- 4 , - 3 , - 2 , - 1 , 0 , + 1 , + 2 , + 3 , + 4	9

Quantum Numbers

Summary

n	l	Orbital shape designation	m_l	Number of orbitals
1	0	1s	0	1
2	0	2s	0	1
	1	2p	- 1 , 0 , + 1	3
3	0	3s	0	1
	1	3p	- 1 , 0 , + 1	3
	2	3d	- 2 , - 1 , 0 , + 1 , + 2	5
4	0	4s	0	1
	1	4p	- 1 , 0 , + 1	3
	2	4d	- 2 , - 1 , 0 , + 1 , + 2	5
	3	4f	- 3 , - 2 , - 1 , 0 , + 1 , + 2 , + 3	7

Nº of subshells: n

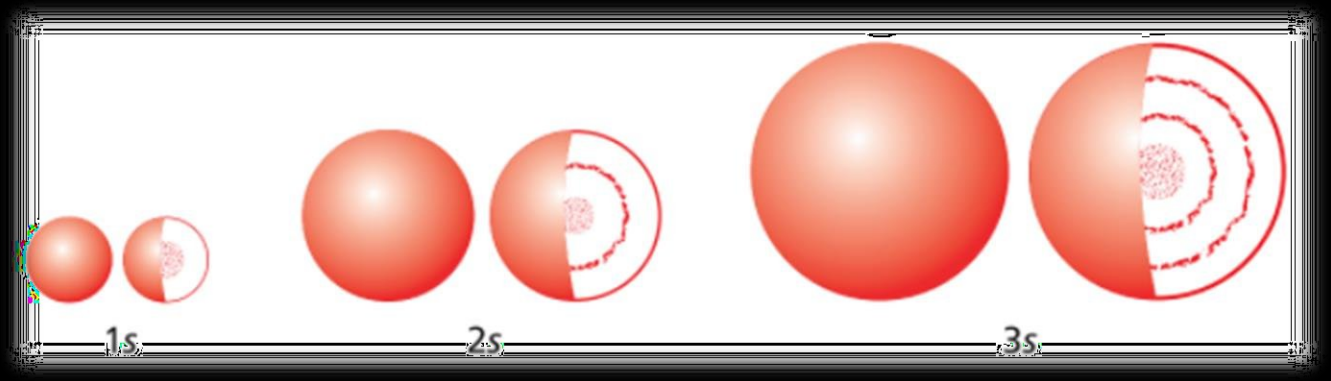
Nº of orbitals within a subshell: 2l+1

Nº of orbitals in a shell: n2

Quantum Numbers

Atomic Orbital's shapes

The shape of the atomic orbitals depend only on **l quantum number**

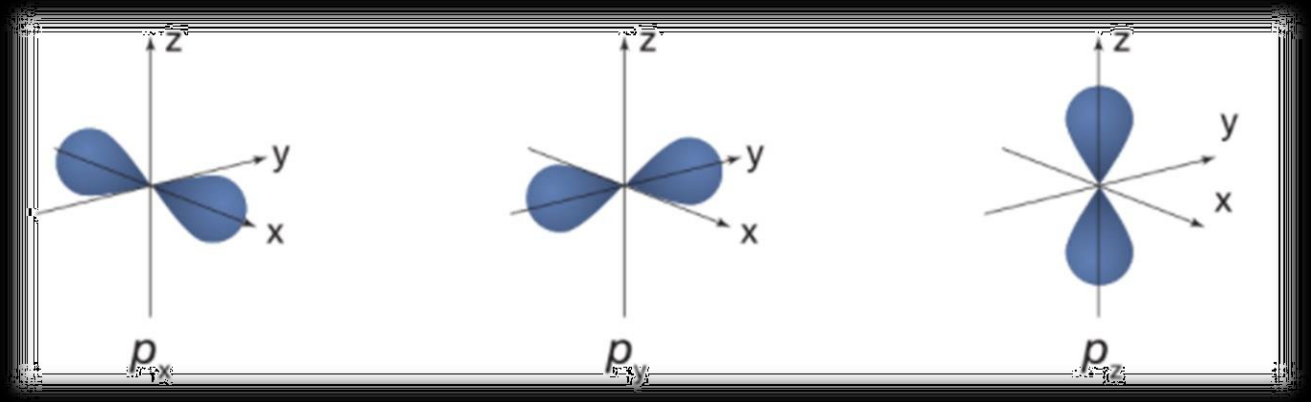


n is " the size " of the atomic orbital.
l value is 0 called s and has a **spherical shape**.
ml value is 0 = no orientation (logic for a sphere)

Quantum Numbers

Atomic Orbital's shapes

The shape of the atomic orbitals depend only on **l quantum number**

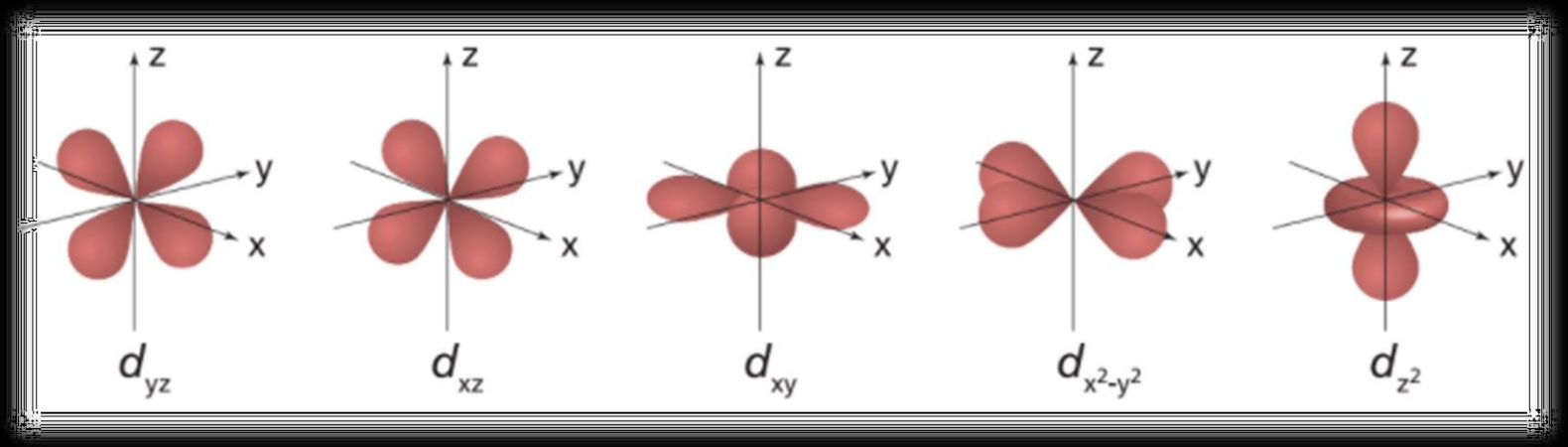


l value is 1 called p and a dumbbell shaped
ml value is 1 = 3 differents orientation (cardinals position x, y z)

Quantum Numbers

Atomic Orbital's shapes

The shape of the atomic orbitals depend only on **l quantum number**



l value is 2 called d and **can take several forms**
ml value is 1 = 5 differents orientation

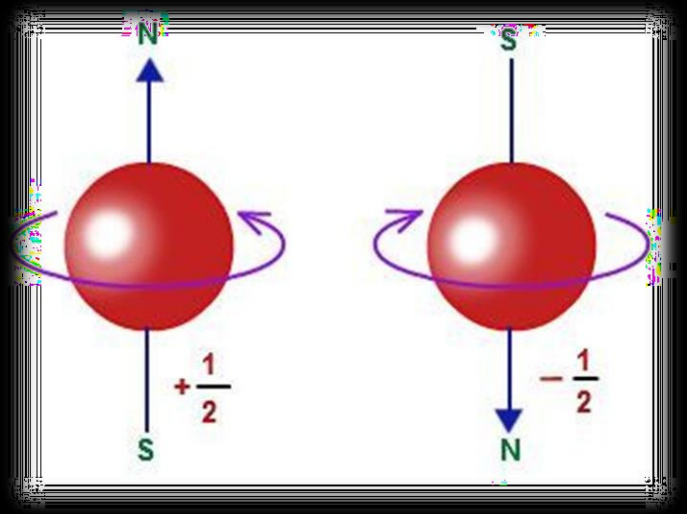
Quantum Numbers

m_s Spin quantum Number

The 3 major quantum numbers as important as they are, couldn't explain some properties observed in 1920s :
Some spectral lines were split in two lines when the theory predicted only one should exist.

A new quantum number was needed to explain the splitting. It is called the Spin Quantum Number

It has only two possible values :+1/2 and -1/2



This image of the 4th quantum number works only when we think the electron as a particle.

Quantum Numbers

Summary 2

- An electron is localized in **atomic orbitals**.
- Each atomic orbital (AO) is characterized by **3 quantum numbers**.
- Each atomic orbital (AO) can contain a **maximum of 2 electrons**.
- The electron **as a spin**.

OVERALL :
One elctron is characterized by 4 quantum

numbers n , l , m_l , m_s

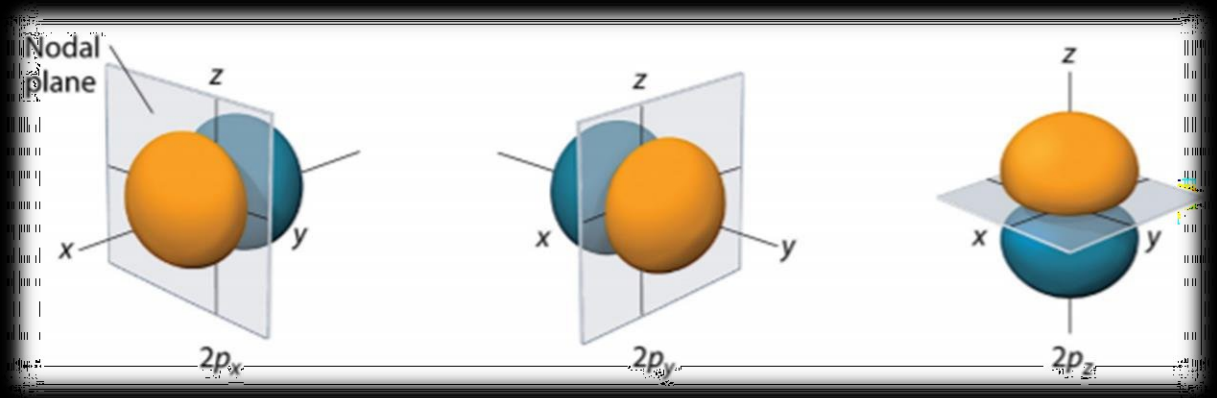
Quantum Numbers

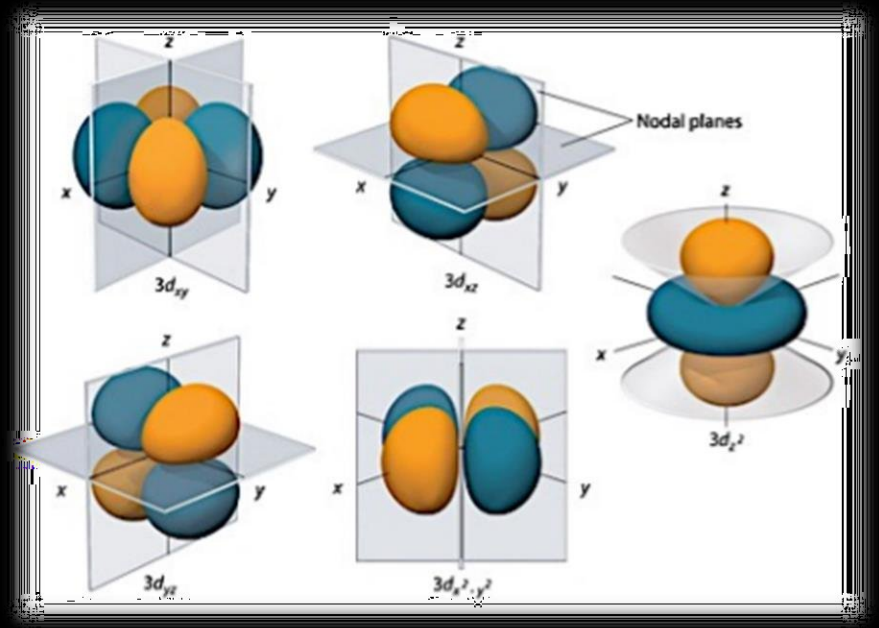
Summary 2

n	# of subshells	Type of subshells	# of orbitals per subshel	# of electrons in each orbital	Electrons in shell
1	1	1s	1	2	2
2	2	2s	1	2	8
		2p	3	6	
3	3	3s	1	2	18
		3p	3	6	
		3d	5	10	
4	4	4s	1	2	32
		4p	3	6	
		4d	5	10	
		4f	7	14	

Atomic Orbitals, Nodal surfaces

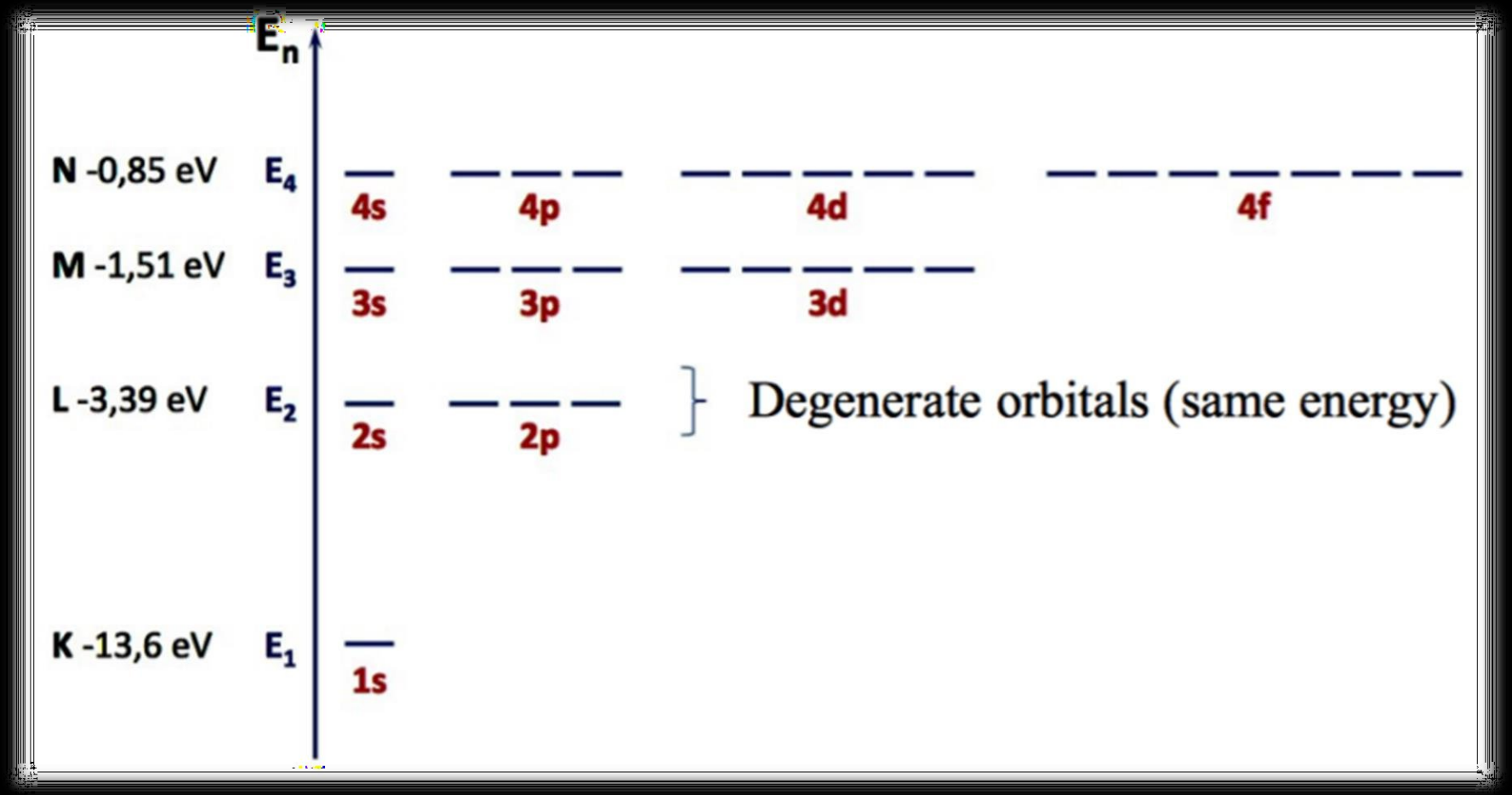
- An Atomic orbital is a probability of presence of an electron.
- Depending on the shape of the AO it exist some region in space where there is no probability to find an electron, called Nodal surfaces





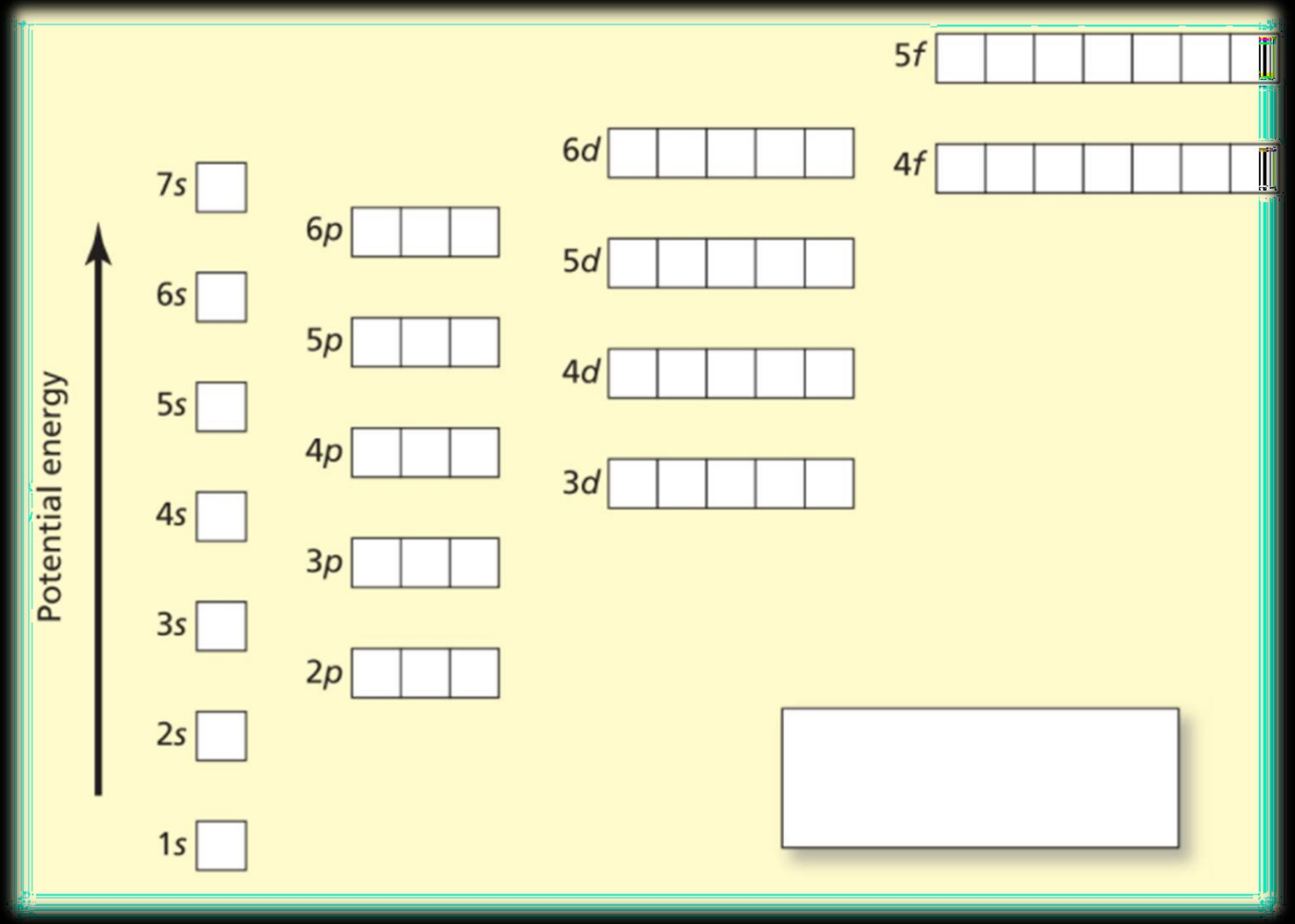
For p and d orbitals: **nodal plane** between de lobes
The orbital **change sign** from one part to the other of the plane

Energetic Diagram



Hydrogen and Hydrogen-like atoms: energy is defined only by quantum number **n**

Energetic Diagram



Many-electron atoms: energy depends on n and l, so each subshell has its own energy

Exercise

One atomic orbital (AO) is characterized by a unique set of _____ quantum numbers

Called _____

One electronis characterized by a unique set of _____ quantum numbers

Called _____

As m_s can have only _____values, one orbital can allocate a maximum of

_____electrons

One orbital is totally occupied when it _____

m_s can take two values : _____and _____

Exercise

How many electrons fit inside the following quantum numbers ?

1. n = 0

2. n = 0
   l = 1

3. n = 1
   ml = 1
   ms = +1/2